

CHAPTER 3

Fundamentals of Temporal Processes

What do we mean by “the past” or “the future” and what does it mean when things “change in time?” We tend to categorize and define events by notions of order, including *past*, *present*, *future*, and *change*. In many respects, it is difficult to imagine a dataset collected in such a way that at least some of these issues are not a factor. The question then is whether the temporal aspect is important to the analysis of the dataset.

Perhaps the easiest way to deal with temporal variability is to assume that it is due to “randomness” and that if enough observations are taken, it will average (in some well defined way) to a central tendency or to a temporal trend. However, it may be that these randomly varying temporal effects are of most interest. Indeed, many environmental, physical, and biological processes are inherently dynamical, and understanding the temporal variability is of primary importance. Statisticians have long been interested in characterizing and modeling such temporal variability, but so has a large community of applied mathematicians, scientists, and engineers.

This chapter provides a very brief introduction to both dynamical systems theory and the statistical analysis of time series. The two areas are related, although the first sometimes takes a deterministic approach and the second a stochastic approach to quantifying temporal variability. Neither is treated comprehensively here, but there is a combined perspective that is very powerful, which we capitalize on in several places in Chapter 7 (where dynamical spatio-temporal models are discussed). By way of a disclaimer, the statistical aspects of time series that we review concentrate on modeling and have less emphasis on inference; for that, there are many excellent texts given in the Bibliographic Notes in Section 3.7.

Traditionally, much of the time series literature has focused on what we describe in Section 2.1 as the “process model.” This is a statistical model

for a (hidden) process, indexed by time, that is either implicitly or explicitly conditioned on underlying parameters that describe the process evolution and/or dependence structure. However, it is also true that there is a tradition in the time series literature of explicitly recognizing that observations come with measurement error. This recognition leads to what are known as *state-space models* in time series. In the terminology of Section 2.1, there is a *data model* (or *measurement equation*, or *observation model*, etc.) that describes the distribution of the data given the underlying process and given parameters that characterize the measurement process. In addition, there is a *process model* (or *state equation*, or *state process*, etc.) that expresses the distribution of the underlying process, conditional on parameters that characterize the process. This framework is inherently hierarchical, and we refer to such models as *hierarchical statistical models*, abbreviated as HMs (Section 2.1).

The hierarchical statistical modeling framework is very powerful and is the focus of much of this book. However, in the context of an introduction to time series, we first discuss process models for temporal (or time series) processes. This will be followed by a more complete discussion of HMs for time series.

3.1 CHARACTERIZATION OF TEMPORAL PROCESSES

In general, a temporal process model can be written as $Y(\cdot)$ or more completely as

$$\{\mathbf{Y}(r) : r \in D_t\}, \quad (3.1)$$

where r indexes the time of the possibly multivariate process $\mathbf{Y}(\cdot)$ and D_t is a subset of $\mathbb{R}^1$. The process $\mathbf{Y}$ may be *deterministic* or *stochastic*. Furthermore, the model is quite general if we allow the possibility that the index set D_t can be a random set.

For a *continuous-time process*, we assume here that D_t is fixed and has nonzero length in the continuous interval $(-\infty, \infty)$, although we often assume $D_t = [0, \infty)$. Similarly, for a *discrete-time process* (time series) we assume a fixed index set of a finite or countable set of times, $D_t = \{0, \pm 1, \pm 2, \dots\}$ (again, we often limit this set to $D_t = \{0, 1, 2, \dots\}$). A third type of process is a *temporal point process*, where D_t is assumed to be a random set made up of randomly occurring points (events) in $\mathbb{R}^1$. For example, the time of occurrence of a tornado in a given county could be represented by a Poisson process in time. We might also be interested in some attribute of the point process, say the severity of the tornado as measured by its maximum wind speed. In this case, we have a *marked temporal point process* in which the mark (or marks) describe some attribute (or attributes) of the process when it occurs.

It is useful to consider the practical distinction between a continuous-time and discrete-time process and, as will be seen below, we choose to notate them differently. In the *physical* world, it is difficult to think of a truly discrete-time

process (except by exogenous aggregation), yet limitations in measurement often force us to consider one. For example, wind speed at a given location in space, considered as a function of time, is an example of a continuous-time process. However, it is not possible to measure such a process continuously, although the temporal resolution can be quite high. Thus, in practice, one might model wind speed as a discrete-time process (e.g., measured hourly). This practical distinction between continuous time and discrete time is common in time series analysis. From a modeling perspective, there are examples where it is important to consider the unobserved (or hidden) process from the continuous-time perspective, even though the observations on that process are taken at discrete times. Many such examples exist in statistical physics and, more recently, in the analysis of financial time series (e.g., Wilmott, Howison, and Dewynne, 1995). In the latter case, if one is interested in trades of some financial instrument at very small time increments, then it makes sense to think about processes with infinitely short time increments, that is, continuous-time processes.

In our notation, we typically refer to a *continuous-time* process with the time index t in parentheses, namely $Y(t)$, and refer to the *discrete-time* process (or time series) with t as a subscript, namely Y_t . In this book, we give a lot of attention to discrete-time processes; but, occasionally, to motivate them, we do present discussion of continuous-time processes.

3.1.1 Joint and Conditional Distributions

Typically, when working with real-world temporal processes, we have only one realization of that process; thus, the associated time series can be viewed as just one sample from a *population*. We characterize this population by its *joint distribution*. For example, consider a time series given by $\{Y_t : t = 0, \dots, T\}$. We denote its joint distribution by

$$[Y_0, Y_1, \dots, Y_T], \quad (3.2)$$

where recall from Chapter 2 that this is short-hand notation used to represent the (joint) probability distribution of $Y_0, Y_1, \dots, Y_T$. This notation is described more completely in Chapter 2, including the conditional probability notation, $[A|B]$. In practice, it can be very difficult to fully specify (3.2) because it requires that we specify all possible interactions between $Y_0, \dots, Y_T$, which can be very problematic in high dimensions. It can often prove useful to think about the joint distribution from a conditional perspective. That is, using basic probability rules (see Chapter 2), we can always rewrite this joint distribution as a product of conditional distributions:

$$[Y_0, \dots, Y_T] = [Y_T|Y_{T-1}, \dots, Y_0][Y_{T-1}|Y_{T-2}, \dots, Y_0] \dots [Y_1|Y_0][Y_0]. \quad (3.3)$$

However, for practical modeling we still need to make additional assumptions about the components of (3.3). For example, we might assume that the process

can be modeled by a first-order Markov property,

$$[Y_t | Y_{t-1}, \dots, Y_0] = [Y_t | Y_{t-1}], \quad \text{for all } t = 1, 2, \dots \quad (3.4)$$

This property suggests a “lack of memory,” so that only the most recent past determines the conditional probabilities about the present given the whole past. In this case, (3.3) becomes

$$[Y_0, \dots, Y_T] = [Y_0] \prod_{t=1}^T [Y_t | Y_{t-1}]. \quad (3.5)$$

The first-order Markov assumption provides a dramatic simplification of the joint distribution, yet the product of these relatively simple conditional distributions can model a process that is quite complicated. Representing complicated joint distributions as a product of conditional distributions is a running theme throughout this book. Although the Markov assumption is very important in time series analysis, there are many other probability models for temporal processes that can be used to simplify joint or conditional distributions.

3.1.2 Deterministic and Stochastic Processes

Although somewhat arbitrary, it is sometimes convenient to characterize temporal processes as either deterministic or stochastic (i.e., random). One might say that for a deterministic process the future is completely determined by the past. For example, a continuous-time process $Y(t)$ might be described via a simple differential equation in which the rate of change of the process Y with time is simply related to a function of the process at time t :

$$\frac{dY(t)}{dt} = f(Y(t)), \quad t \geq 0, \quad (3.6)$$

where the function f may be linear or nonlinear in $Y(t)$. Thus, given some *initial condition* $Y(0)$, the evolution of the process is completely determined by the function f . Alternatively, if we limit our discussion to only processes at discrete times, then we could write an analogous equation,

$$Y_t = \mathcal{M}(Y_{t-1}), \quad t = 1, 2, \dots, \quad (3.7)$$

where the function $\mathcal{M}$ maps the process from the previous time $t - 1$ to the current time t . Again, given an initial condition Y_0 , the process $\{Y_t : t = 0, 1, 2, \dots\}$ is completely determined by the function $\mathcal{M}$.

For *stochastic* or *random* processes, we might say that the future is only partially determined from the past. For example, we can add a random “noise” component to equation (3.7):

$$Y_t = \mathcal{M}(Y_{t-1}) + \eta_t, \quad t = 1, 2, \dots, \quad (3.8)$$

where $\{\eta_t : t = 1, 2, \dots\}$ is a mean-zero random process and η_t is statistically independent of Y_{t-1} . This random-noise term implies that the process $\{Y_t\}$ is random as well. Similarly, stochastic differential equations can be formed by the addition of a continuous-time random-noise term to (3.6) (e.g., see the introduction to Chapter 6). The process $\{Y_t : t = 0, 1, \dots\}$ in (3.8) is an example of what could generally be called a *discrete-time temporal random process*.

Just as the distinction between continuous and discrete time is not always clear, so it is with the difference between deterministic and stochastic processes. For example, consider a *chaotic* process, which is a nonlinear, deterministic temporal process that exhibits sensitivity to initial conditions (e.g., see Section 3.2.4; and Devaney, 2003, p. 50). That is, for nonlinear functions f in (3.6) (or $\mathcal{M}$ in (3.7)), very small changes to the initial condition $Y(0)$ can lead to very different trajectories of the process $Y(t)$. Thus, even though such trajectories are completely determined once the initial condition is known, two slightly different initial conditions can ultimately exhibit sequences that do not appear to be at all related. The practical implication of this is that since there is always some uncertainty in initial conditions for real-world processes (e.g., due to measurement error), nonlinear deterministic processes will ultimately become unpredictable. In fact, when there is this initial-condition uncertainty, one cannot typically distinguish between chaotic uncertainties and uncertainty due to a random process (e.g., see Berliner, 1992; Chan and Tong, 2001; and Section 3.2.2, for more discussion).

The next section gives a brief introduction to *deterministic* dynamical systems. The remainder of the chapter then focuses on random temporal processes.

3.2 INTRODUCTION TO DETERMINISTIC DYNAMICAL SYSTEMS

To facilitate the discussion of statistical models for spatio-temporal dynamical processes, it is necessary to provide some background to deterministic dynamical systems. This discussion is not intended to be a formal treatment of dynamical-systems theory, but rather it is an attempt to provide some definitions and basic concepts in the area. For a detailed and technical overview, the reader is referred to the references in the Bibliographic Notes in Section 3.7.

We say the *state* of a system at time t is described by a set of p_α variables, $\alpha_t = (\alpha_t(1), \dots, \alpha_t(p_\alpha))'$, which are sometimes known as the *state variables* or, collectively, as the *state process*. In general, state variables may be physical variables (e.g., temperature, pressure, wind components, water vapor, etc., in the atmosphere) defined at specific locations in physical space. From our perspective, any system whose state changes with time is a *dynamical system*. The evolution of the system through *state space* (or, equivalently, *phase space*) is often called a *trajectory*. If the current state of the system, say α_t , uniquely determines the future states $\alpha_{t+\tau}$; $\tau > 0$, the system is referred to as

a *deterministic dynamical system*. Correspondingly, if such a unique mapping does not exist, we say the system is a *stochastic dynamical system*.

In general, one might consider discrete-time or continuous-time dynamical systems. In this section, we focus on discrete systems of the form

$$\alpha_t = \mathcal{M}(\alpha_{t-1}; \theta), \quad t = 1, 2, \dots, \quad (3.9)$$

where the *mapping* (or *transition*) function $\mathcal{M}$, describes the behavior of the state process as time evolves, and θ is a vector of parameters involved in that description. To complete the definition of the dynamical system, one needs to specify the initial condition, α_0 , as well as boundary conditions if the elements of α_t depend on boundary values. In the case where the mapping $\mathcal{M}$ varies with time, the system is said to be *nonautonomous*, and where $\mathcal{M}$ does not vary with time, it is said to be *autonomous*. A critical distinction is made between systems in which $\mathcal{M}$ is linear (in the state variables) versus those in which $\mathcal{M}$ is nonlinear. We first consider the case where $\mathcal{M}$ is linear.

3.2.1 Linear Discrete Dynamical Systems

Our discussion of linear discrete dynamical systems follows the introductory treatment in Sandefur (1990).

First-Order Univariate Systems

Initially, we consider the first-order *linear* discrete dynamical system. In general, a univariate first-order system might be written as

$$\alpha_t = \mathcal{M}(\alpha_{t-1}; \theta), \quad t = 1, 2, \dots, \quad (3.10)$$

and when $\mathcal{M}$ is linear, it becomes a *first-order affine dynamical system*:

$$\alpha_t = \theta_0 + \theta_1 \alpha_{t-1}, \quad t = 1, 2, \dots, \quad (3.11)$$

where θ_0 and θ_1 are parameters. Even such a simple system is useful for establishing some basic definitions and concepts. For example, we say an *equilibrium value* or *fixed point* of the system is a number “ a ” such that $\alpha_t = a$ for all values of t when $\alpha_0 = a$. That is, for the general formulation (3.10), a is an equilibrium value if and only if $\mathcal{M}(a) = a$. Thus, for the linear system (3.11), the equilibrium value is $a = \theta_0 / (1 - \theta_1)$ when $\theta_1 \neq 1$ (if $\theta_1 = 1$, then there is no equilibrium value). Of course, in general, a linear dynamical system can have many different equilibrium values.

Example: Consider a simple example where there is an initial animal population of size α_0 and it is assumed to grow by 5 percent per year. In addition, 30 members of the population are removed each year to supplement smaller

populations at other (distinct) locations. A simple linear dynamical system describes how the population varies each year:

$$\alpha_t = 1.05\alpha_{t-1} - 30. \quad (3.12)$$

In this case, the fixed point is $a = 600$; thus, if $\alpha_0 = 600$, the population will stay constant through time ($\alpha_1 = 600, \alpha_2 = 600, \dots$). If the initial population is fewer than 600, the population will decrease with time, and if the initial population is more than 600, the population will increase with time.

With regard to (3.10), it is interesting to consider what happens as the state of the system nears the equilibrium value. Intuitively, an equilibrium value is said to be *stable* or *attracting* if α_0 “close to” a implies that α_t tends to a (in a limiting sense). The equilibrium value is said to be *unstable* or *repelling* if, regardless of how close the initial condition α_0 is to a , α_t eventually ends up “far from” a (note that for an unstable equilibrium, the trajectory can come close to a but it cannot stay close). For example, the equilibrium value for the first-order system (3.11) is stable if $|\theta_1| < 1$. However, for $|\theta_1| > 1$, the equilibrium value is unstable ($|\alpha_t|$ tends to infinity for $\alpha_0 \neq a$). For the special case when $\theta_1 = -1$, we see that $\alpha_0 = \alpha_2 = \alpha_4 = \dots$ and $\alpha_1 = \alpha_3 = \alpha_5 = \dots$. This type of pattern, often seen in dynamical systems, is known as a *2-cycle*. In this case, since α_t neither converges nor diverges from the equilibrium value, the equilibrium value is said to be *neutral*.

In simple linear systems, for $t = 0, 1, \dots$, we can obtain solutions of α_t . Consider the first-order dynamical system given in (3.11) but with $\theta_0 = 0$. It is easy to show that the general solution to this system is $\alpha_t = c\theta_1^t$, for $t = 0, 1, \dots$, where c is a constant. Analogous to differential equations, the particular solution can be found, given α_0 , to be $\alpha_t = \alpha_0\theta_1^t$, for $t = 0, 1, \dots$. The more general affine system (with an additive constant θ_0) given in (3.11) has general solution $\alpha_t = c\theta_1^t + \theta_0/(1 - \theta_1)$, provided $\theta_1 \neq 1$ and for $t = 0, 1, \dots$. The corresponding particular solution, given α_0 , can be shown to be: $\alpha_t = \theta_1^t(\alpha_0 - \theta_0/(1 - \theta_1)) + \theta_0/(1 - \theta_1)$, for $t = 0, 1, \dots$. The following properties of the solution to the first-order system are useful for describing system behavior:

- If $0 < \theta_1 < 1$, then $\alpha_t \rightarrow a$, exponentially.
- If $-1 < \theta_1 < 0$, then α_t oscillates in a dampened manner to a .
- If $|\theta_1| > 1$, then $\alpha_t \rightarrow \pm\infty$, exponentially fast; and if $\theta_1 < -1$, then α_t oscillates with increasing amplitude.

Note, for the special case where $\theta_1 = 1$, the general solution is $\alpha_t = c + \theta_0 t$, and if the initial condition is given, the particular solution is, $\alpha_t = \alpha_0 + \theta_0 t$.

Example: Consider the simple system, $\alpha_t = -.7\alpha_{t-1}$, and $\alpha_0 = 12$. Figure 3.1 shows realizations from this system through time $t = 10$. Given

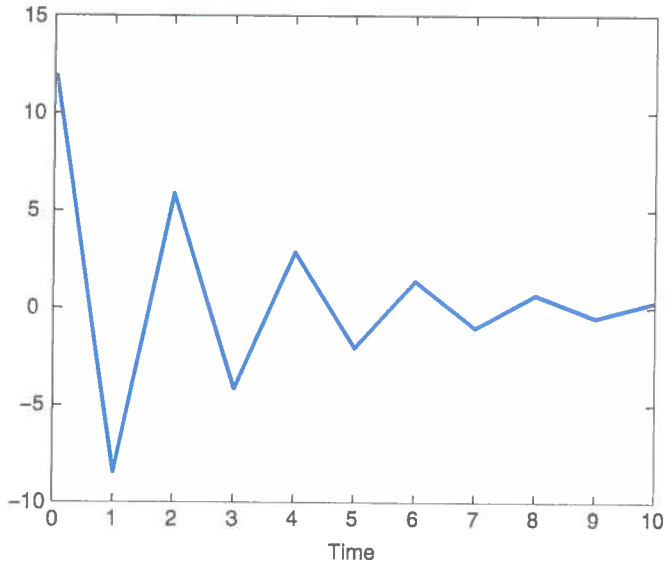


Figure 3.1 The first-order dynamical system: $\alpha_t = -0.7\alpha_{t-1}$, where $\alpha_0 = 12$.

that $-1 < \theta_1 < 0$, we see that the solution is oscillating and decreasing in amplitude over time.

A thorough study of linear dynamical systems considers nonhomogeneous systems such as

$$\alpha_t = \theta_1 \alpha_{t-1} + g_t, \quad t = 1, 2, \dots, \quad (3.13)$$

where typically $\{g_t : t = 0, 1, \dots\}$ is some deterministic function of t . Such systems do not have equilibrium points. Solutions can be found for specific functional forms for g_t (e.g., when g_t is a polynomial in t or when g_t is an exponential in t). We do not present such systems here, but refer the interested reader to Sandefur (1990).

Second-Order Univariate Systems

The second-order linear discrete homogeneous dynamical system is given by

$$\alpha_t = \theta_1 \alpha_{t-1} + \theta_2 \alpha_{t-2}, \quad t = 2, 3, \dots \quad (3.14)$$

It is convenient at this point to introduce some notation. We define the *backward-shift operator* B as follows:

$$\alpha_{t-1} = B\alpha_t. \quad (3.15)$$

Hence,

$$\alpha_{t-2} = B\alpha_{t-1} = BB\alpha_t = B^2\alpha_t, \quad (3.16)$$

and in general,

$$\alpha_{t-p} = B^p\alpha_t. \quad (3.17)$$

Although not considered here, we note that one could use analogous “forward-shift operators” in this context if the general dynamical system is written as $\alpha_{t+1} = \mathcal{M}(\alpha_t; \theta)$. Using the backward-shift notation, we can rewrite the dynamical system (3.14) as

$$(1 - \theta_1 B - \theta_2 B^2)\alpha_t = 0, \quad t = 2, 3, \dots, \quad (3.18)$$

and we let $1 - \theta_1 z - \theta_2 z^2$ be the *characteristic polynomial* of this system. The behavior of the dynamical system is described by the roots of this polynomial, which we denote by z_1 and z_2 . We consider two general cases.

Case 1: $z_1 \neq z_2$. In this case, the general solution of the second-order system is

$$\alpha_t = c_1 z_1^{-t} + c_2 z_2^{-t}, \quad t = 0, 1, \dots, \quad (3.19)$$

where the constants c_1 and c_2 depend on the initial conditions, α_0 and α_1 :

$$c_1 = \frac{\alpha_1 - \alpha_0 z_2^{-1}}{z_1^{-1} - z_2^{-1}}, \quad c_2 = \frac{\alpha_1 - \alpha_0 z_1^{-1}}{z_2^{-1} - z_1^{-1}}. \quad (3.20)$$

Before we proceed, it is helpful to recall the following facts concerning complex numbers and sinusoidal processes. Let $h = u + iv$ be a complex number, where $i \equiv \sqrt{-1}$. The complex conjugate is $\bar{h} = u - iv$, and the modulus is $|h| = \sqrt{u^2 + v^2}$. On occasion, notation is simplified by the *Euler formula*, $\exp(i\gamma) = \cos(\gamma) + i \sin(\gamma)$, or, $\exp(-i\gamma) = \cos(\gamma) - i \sin(\gamma)$. In addition, the *complex plane* is such that the real component (u) is plotted along the horizontal axis and the imaginary component (v) is plotted along the vertical axis. The resulting vector has magnitude given by the modulus $|h|$, and the angle γ between the imaginary and real components is defined such that $\tan(\gamma) = v/u$; thus, $u = |h| \cos(\gamma)$, $v = |h| \sin(\gamma)$, and $h = |h| \exp(i\gamma)$.

In addition, consider a sinusoidal process that is defined in terms of its *amplitude*, *frequency*, and *phase*:

$$Y_t = A \cos(\omega t + \gamma), \quad (3.21)$$

where A is the amplitude, ω is the frequency (in units of radians per time unit), and γ is the phase (in radians). We sometimes refer to γ also as the phase shift.

Using the trigonometric identity, $\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$ we can rewrite (3.21) as

$$\begin{aligned} Y_t &= A \cos(\gamma) \cos(\omega t) + (-A \sin(\gamma)) \sin(\omega t) \\ &= a \cos(\omega t) + b \sin(\omega t), \end{aligned} \quad (3.22)$$

where $a = A \cos(\gamma)$ and $b = -A \sin(\gamma)$. Thus, if a and b are known, then the amplitude is given by $A = \sqrt{a^2 + b^2}$ and the phase is given by $\gamma = \tan^{-1}(-b/a)$.

Now, for the special case where the two roots, z_1 and z_2 , are a complex-conjugate pair, the general solution is still given by (3.19). However, since $z_2 = \bar{z}_1$ and, necessarily, $c_2 = \bar{c}_1$ (since the solution is real), we can write

$$\alpha_t = c_1 z_1^{-t} + \bar{c}_1 \bar{z}_1^{-t}. \quad (3.23)$$

In terms of complex numbers, the components of (3.23) can be written as

$$c_1 = |c_1| \exp(i\psi), \quad z_1 = |z_1| \exp(i\phi), \quad (3.24)$$

where ψ is the angle between the real and imaginary components of c_1 , and ϕ is the angle between the real and imaginary components of z_1 . We can write analogous complex notation for the complex conjugates, $\bar{c}_1$ and $\bar{z}_1$. Then the general solution becomes

$$\alpha_t = 2|c_1||z_1|^{-t} \cos(t\phi - \psi), \quad (3.25)$$

where the constants c_1 and ψ must be determined by the initial conditions, α_0 and α_1 . The important point to note regarding (3.25) is that the solution exhibits a sinusoidal oscillation with amplitude depending on t and given by $2|c_1||z_1|^{-t}$, frequency given by ϕ , and phase shift given by ψ .

Case 2: $z_1 = z_2 (= z_0)$. In this case, the general solution of the second-order system is

$$\alpha_t = (c_1 + c_2 t) z_0^{-t}, \quad k = 0, 1, \dots, \quad (3.26)$$

where the constants c_1 and c_2 depend on the initial conditions, α_0 and α_1 :

$$c_1 = \alpha_0, \quad c_2 = \alpha_1 z_0 - \alpha_0. \quad (3.27)$$

Not surprisingly, the behavior of the second-order system relative to its equilibrium value depends on the nature of the roots, z_1 and z_2 . First, consider the second-order affine system:

$$\alpha_t = \theta_1 \alpha_{t-1} + \theta_2 \alpha_{t-2} + \theta_0, \quad t = 2, 3, \dots \quad (3.28)$$

The equilibrium value in this case is given by

$$a = \frac{\theta_0}{1 - \theta_1 - \theta_2}, \quad (3.29)$$

provided $\theta_1 + \theta_2 \neq 1$. In the case where z_1 and z_2 are distinct and not equal to 1, the general solution of (3.28) is given by $\alpha_t = c_1 z_1^{-t} + c_2 z_2^{-t} + a$. One can show that if $|z_1| > 1$ and $|z_2| > 1$, then the equilibrium value is attracting (stable). Furthermore, if either $|z_1| < 1$ or $|z_2| < 1$, or both, then the equilibrium value is repelling (unstable).

Example: Consider the dynamical system:

$$\alpha_t = -0.64\alpha_{t-2}, \quad t = 2, 3, \dots, \quad (3.30)$$

with initial conditions $\alpha_0 = 0$ and $\alpha_1 = -5$. The roots of the polynomial, $0.64z^2 + 1 = 0$, are $z_1 = 1.25i$ and $z_2 = -1.25i$. In this case, the general solution is given by (3.25). We also note that $|z_1| = |z_2| = 1.25$, and so the system is stable. In addition, $\phi = \pi/2$ and, given the initial conditions, we find that $\psi = \pi/2$ and $|c_1| = 3.125$, resulting in the solution, $\alpha_t = 6.25(1.25)^{-t} \cos(t\pi/2 + \pi/2)$. This system is shown in Figure 3.2 for time $t = 0, 1, \dots, 20$, where the damped oscillation is clearly evident.

Linear Dynamical Systems of Several Variables

Consider the p_α -dimensional state process $\{\alpha_t\}$, given by $\alpha_t = (\alpha_t(1), \dots, \alpha_t(p_\alpha))'$. For example, $\alpha_t(j)$ might represent the temperature at the j th spatial

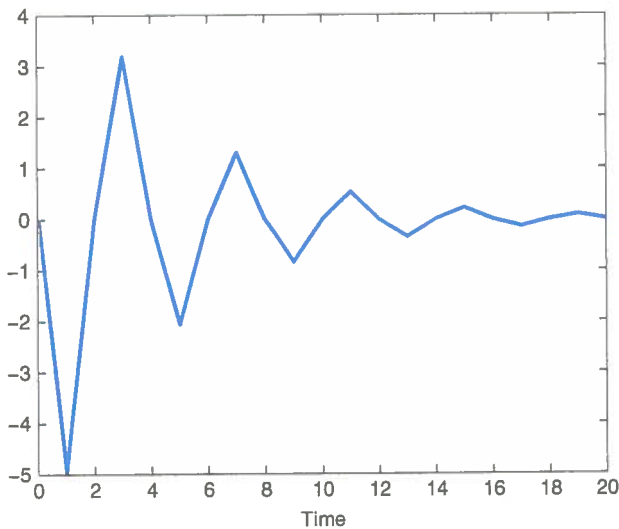


Figure 3.2 The second-order dynamical system: $\alpha_t = -0.64\alpha_{t-2}$, where $\alpha_0 = 0$, $\alpha_1 = -5$.

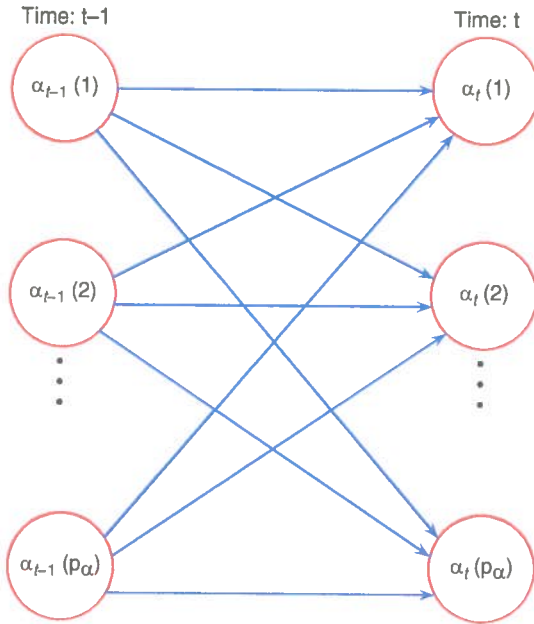


Figure 3.3 Schematic of a p_α -dimensional dynamical system in which each component at time t can be influenced by each component at the previous time, $t - 1$.

location at time t , or the population of the j th species at time t , or the abundance of the j th age class of a species at time t , and so on. Our goal is to describe a system in which the components of this state process interact across time, as shown schematically in Figure 3.3.

In a first-order system, we would have

$$\alpha_t(j) = m_{j1}\alpha_{t-1}(1) + m_{j2}\alpha_{t-1}(2) + \cdots + m_{jp_\alpha}\alpha_{t-1}(p_\alpha), \quad (3.31)$$

for $j = 1, \dots, p_\alpha$. Thus, $\alpha_t(j)$ is related to itself and the other components of the state process at the previous time, according to the transition coefficients, m_{jk} , $k = 1, \dots, p_\alpha$. Obviously, the notation will simplify if we write this in an equivalent matrix form:

$$\alpha_t = \mathbf{M}\alpha_{t-1}, \quad t = 1, 2, \dots, \quad (3.32)$$

where

$$\mathbf{M} \equiv \begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1p_\alpha} \\ m_{21} & \cdots & & m_{2p_\alpha} \\ \vdots & & & \vdots \\ m_{p_\alpha 1} & \cdots & & m_{p_\alpha p_\alpha} \end{pmatrix}. \quad (3.33)$$

By substitution, the general solution of this system is given by

$$\alpha_t = \mathbf{M}'^t \alpha_0, \quad t = 0, 1, \dots, \quad (3.34)$$

where $\mathbf{M}' \equiv \mathbf{M}\mathbf{M} \dots \mathbf{M}$ (t products of the matrix $\mathbf{M}$), and a particular solution is obtained if one specifies the starting value, α_0 .

One might ask just how general the first-order vector ("multivariate") dynamical system is? In fact, it can be quite general. As an example, consider the second-order univariate system,

$$\beta_t = b_1 \beta_{t-1} + b_2 \beta_{t-2}, \quad t = 2, 3, \dots \quad (3.35)$$

We can rewrite this as a system of equations by defining $\alpha_{t-1} = (\alpha_{t-1}(1), \alpha_{t-1}(2))'$, where $\alpha_{t-1}(1) \equiv \beta_{t-1}$, $\alpha_{t-1}(2) \equiv \beta_{t-2}$. Then, the system (3.35) can be rewritten as

$$\alpha_t = \mathbf{M} \alpha_{t-1}, \quad (3.36)$$

where

$$\mathbf{M} = \begin{pmatrix} b_1 & b_2 \\ 1 & 0 \end{pmatrix}. \quad (3.37)$$

Note that such a reparameterization approach easily extends to higher-order vector systems. For example, for vectors $\{\beta_t\}$, consider the third-order system,

$$\beta_t = \mathbf{G}_1 \beta_{t-1} + \mathbf{G}_2 \beta_{t-2} + \mathbf{G}_3 \beta_{t-3}. \quad (3.38)$$

Then define $\alpha_t \equiv (\beta_t', \beta_{t-1}', \beta_{t-2}')'$, resulting in

$$\alpha_t = \begin{pmatrix} \mathbf{G}_1 & \mathbf{G}_2 & \mathbf{G}_3 \\ \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} \end{pmatrix} \alpha_{t-1} \equiv \mathbf{M} \alpha_{t-1}. \quad (3.39)$$

Thus, the first-order vector dynamical system is quite general.

Spectral Representation of the First-Order System

Consider the first-order system (3.32). In general, the matrix $\mathbf{M}$ is nonsymmetric. As shown, for example, by Rao (1965, pp. 38–39), the characteristic equation, $|\mathbf{M} - \lambda \mathbf{I}| = 0$, has p_α roots, some of which may be complex, even if $\mathbf{M}$ is real. Thus, corresponding to each *latent root*, λ_i , there are two vectors, $\mathbf{w}_i$ and $\mathbf{v}_i$, called the right and left *singular vectors*, respectively, such that

$$\mathbf{M} \mathbf{w}_i = \lambda_i \mathbf{w}_i, \quad (3.40)$$

$$\mathbf{M}' \mathbf{v}_i = \lambda_i \mathbf{v}_i, \quad (3.41)$$

for $i = 1, \dots, p_\alpha$. Then, the following results can be obtained (e.g., Rao, 1965, pp. 38–39):

- The right eigenvectors $\{\mathbf{w}_i\}$ are linearly independent and so are the left eigenvectors $\{\mathbf{v}_i\}$. This implies that $\mathbf{W} \equiv (\mathbf{w}_1, \dots, \mathbf{w}_{p_\alpha})$ and $\mathbf{V} \equiv (\mathbf{v}_1, \dots, \mathbf{v}_{p_\alpha})$ are nonsingular; thus,

$$\mathbf{M} = \mathbf{W}\mathbf{A}\mathbf{W}^{-1} \quad \text{and} \quad \mathbf{M}' = \mathbf{V}\mathbf{A}\mathbf{V}^{-1}, \quad (3.42)$$

where $\mathbf{A} \equiv \text{diag}(\lambda_1, \dots, \lambda_{p_\alpha})$.

- Orthogonality is of the form, $\mathbf{w}'_i \mathbf{v}_j = 0$ for $i \neq j$ and $\mathbf{w}'_i \mathbf{v}_i \neq 0$ for $i = 1, \dots, p_\alpha$.
- Since $\mathbf{M} = \mathbf{W}\mathbf{A}\mathbf{W}^{-1}$, then $\mathbf{W}^{-1}\mathbf{M} = \mathbf{A}\mathbf{W}^{-1}$, which implies that $\mathbf{W}^{-1} = \mathbf{V}'$; thus, $\mathbf{M} = \mathbf{W}\mathbf{A}\mathbf{V}' = \sum_{i=1}^{p_\alpha} \lambda_i \mathbf{w}_i \mathbf{v}'_i$. *Note:* This assumes that $\mathbf{W}\mathbf{V}' = \mathbf{I}$ (i.e., that the latent roots are normalized). If, in general, $\mathbf{W}\mathbf{V}' = \mathbf{D}$ (where $\mathbf{D}$ is a diagonal matrix, and the singular vectors are chosen so that its diagonal elements are positive), then $\mathbf{M} = \sum_{i=1}^{p_\alpha} (\lambda_i/d_i) \mathbf{w}_i \mathbf{v}'_i$, where d_i is the i th diagonal element of $\mathbf{D}$.

We can use these results to express the solution to the general first-order difference equation in terms of its spectral components $\{(\lambda_i, \mathbf{w}_i, \mathbf{v}_i) : i = 1, \dots, p_\alpha\}$. Using the orthogonality result, one can see that $\mathbf{M}^2 = \mathbf{W}\mathbf{A}\mathbf{V}'\mathbf{W}\mathbf{A}\mathbf{V}' = \mathbf{W}\mathbf{A}^2\mathbf{V}'$. Furthermore, in general, $\mathbf{M}^k = \mathbf{W}\mathbf{A}^k\mathbf{V}'$; $k = 1, 2, \dots$. Thus, we can write the general solution,

$$\boldsymbol{\alpha}_t = \mathbf{M}'^t \boldsymbol{\alpha}_0 = \mathbf{W}\mathbf{A}'^t \mathbf{V}' \boldsymbol{\alpha}_0 = \sum_{i=1}^{p_\alpha} \lambda'_i \mathbf{w}_i \mathbf{v}'_i \boldsymbol{\alpha}_0, \quad t = 0, 1, \dots \quad (3.43)$$

Suppose that because of the linear independence of $\{\mathbf{w}_i\}$, we express the initial condition $\boldsymbol{\alpha}_0 = \mathbf{W}\mathbf{c} = \sum_{i=1}^{p_\alpha} c_i \mathbf{w}_i$, for some constants $\mathbf{c} \equiv (c_1, \dots, c_{p_\alpha})'$. Then, it follows that $\mathbf{c} = \mathbf{W}^{-1} \boldsymbol{\alpha}_0 = \mathbf{V}' \boldsymbol{\alpha}_0$, and the particular solution can be written as

$$\boldsymbol{\alpha}_t = \mathbf{W}\mathbf{A}'^t \mathbf{c} = \sum_{i=1}^{p_\alpha} c_i \lambda'_i \mathbf{w}_i, \quad t = 0, 1, \dots \quad (3.44)$$

Now, it is clear that the solution to the linear difference equation is critically dependent on the values of $\{\lambda_i\}$. For example, if $|\lambda_i| < 1$, then $\lim_{t \rightarrow \infty} \lambda'_i = 0$, and thus the i th term in the linear expansion of $\boldsymbol{\alpha}_t$ goes to zero. More generally, let $\|\mathbf{M}\| = \max\{|\lambda_i| : i = 1, \dots, p_\alpha\}$. Then, if $\|\mathbf{M}\| < 1$, it follows that $\lim_{t \rightarrow \infty} \boldsymbol{\alpha}_t = \mathbf{0}$. In this case, the zero vector is a stable equilibrium value. In general, one can show the following:

- If λ_i is real and $\lambda_i > 1$, then the i th component of $\boldsymbol{\alpha}_t$ in the expansion (3.44) exhibits exponential growth.
- If λ_i is real and $0 < \lambda_i < 1$, then the i th component of $\boldsymbol{\alpha}_t$ in the expansion (3.44) exhibits exponential decay.

- If λ_i is real and $-1 < \lambda_i < 0$, then the i th component of α_t in the expansion (3.44) exhibits a damped oscillation with period 2.
- If λ_i is real and $\lambda_i < -1$, then the i th component of α_t in the expansion (3.44) exhibits a divergent oscillation with period 2.
- If λ_i is complex, then there is an associated complex conjugate component in the expansion (3.44). In this case, the associated components of α_t in the expansion (3.44) exhibit oscillations as in the second-order dynamical system (see Section 3.2.1). That is, suppose $\lambda_i = a + b\sqrt{-1}$. Then, using common identities for complex numbers, $\lambda_i^t = |\lambda_i|^t (\cos \phi t + \sqrt{-1} \sin \phi t)$, where $|\lambda_i| = \sqrt{a^2 + b^2}$ is the amplitude and $\phi = \tan^{-1}(b/a)$ is the phase angle formed by λ_i in the complex plane. For a real solution, we must have complex conjugate pairs of eigenvalues, and so

$$\lambda_i^t + \bar{\lambda}_i^t = 2|\lambda_i|^t \cos(\phi t). \quad (3.45)$$

Thus, the amplitude of the oscillation increases or decreases, depending on whether $|\lambda_i|$ is greater or less than 1, respectively. For each unit increase in time, the angle in the complex plane increases by ϕ , and an oscillation is completed with period $2\pi/\phi$.

- If $\lambda_i = 1$ and $|\lambda_j| < 1$, for all $j \neq i$, then $\lim_{t \rightarrow \infty} \alpha_t = c_i \mathbf{w}_i$, where the constant c_i depends on the initial condition. We note that this solution is not a stable equilibrium since, depending on the initial condition, the solution will be any multiple of $\mathbf{w}_i$.
- If $\lambda_i = -1$ and $|\lambda_j| < 1$, for all $j \neq i$, then the limiting solution is $\alpha_t = c_i(-1)^t \mathbf{w}_i$, which is a 2-cycle, with the constant c_i determined from the initial condition.
- If $\lambda_i = 1$, $\lambda_j = -1$, $|\lambda_l| < 1$, for all $l \neq i, j$, then the limiting solution is again a 2-cycle: $\alpha_t = c_i \mathbf{w}_i + c_j(-1)^t \mathbf{w}_j$, with c_i, c_j determined by the initial condition.
- If $\lambda_i = a + b\sqrt{-1}$ and $\lambda_j = a - b\sqrt{-1}$, with $a^2 + b^2 = 1$, and $|\lambda_l| < 1$, $l \neq i, j$, then the limiting solution is a constant amplitude oscillation.

It is often useful to examine the sensitivity of the latent roots to changes in the matrix elements m_{qr} through a *sensitivity matrix* (e.g., see Caswell, 2001, Section 9.1). That is, the sensitivity matrix associated with the i th latent root is

$$\mathbf{S}_i \equiv \left(\frac{\partial \lambda_i}{\partial m_{qr}} \right) = \frac{\bar{\mathbf{v}}_i \mathbf{w}'_i}{\mathbf{v}'_i \mathbf{w}_i}. \quad (3.46)$$

Furthermore, given that the latent roots of $\mathbf{M}$ reveal so much about the underlying dynamics of the system, it is often useful to consider the *eigenvalue spectrum* of $\mathbf{M}$, as illustrated below.

Table 3.1 Eigenvalue Spectrum of $\mathbf{M}$

λ_i	$ \lambda_i $	ϕ/π
1.3070	1.3070	0
$0.8465 + 0.2840\sqrt{-1}$	0.8929	0.1030
$0.8465 - 0.2840\sqrt{-1}$	0.8929	-0.1030

Example: Consider the case where $p_\alpha = 3$ and $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{pmatrix} 1.1 & 0.3 & 0.2 \\ 0.2 & 1.0 & -0.1 \\ -0.2 & 0.4 & 0.9 \end{pmatrix}. \quad (3.47)$$

Then the eigenvalue spectrum is given in Table 3.1.

We can see from this table that there is a dominant mode with $|\lambda_1| = 1.3070 > 1$, implying that α_t will exhibit exponential growth over time. In addition, assuming the time increment is one year, there is a decaying oscillation with period $2\pi/\phi = 19.4$ years. As time increases, this periodic component is dominated by the exponentially growing mode.

The sensitivity matrix for λ_1 in this case is

$$\mathbf{S}_1 = \begin{pmatrix} 0.56 & 0.35 & 0.06 \\ 0.69 & 0.42 & 0.08 \\ 0.11 & 0.07 & 0.01 \end{pmatrix}, \quad (3.48)$$

suggesting that the value of λ_1 is most sensitive to m_{21} , the element of $\mathbf{M}$ in the second row and first column.

The eigenvalue spectrum can be very useful across a wide variety of applications. For example, in ecology and demography, researchers consider so-called “matrix models” [see Caswell (2001)], which are simply discrete linear systems of the structure considered here, where, for example, the components of the state vector might be the number of organisms in different age classes. The transition matrix in that setting then relates the flow of members of the population from one class to another. Another application, more relevant to the spatio-temporal setting of interest here, is to the so-called Principal Oscillation Pattern (POP) analysis. Such analysis considers a spatial process that evolves in time according to a first-order system, such as one might find in meteorology or oceanography. This is considered in detail in Section 5.5.

3.2.2 Nonlinear Discrete Dynamical Systems

Many of the basic concepts defined for linear systems also hold for nonlinear systems, but some modifications and additional concepts must be introduced

to describe the more complicated behavior of nonlinear systems. Again, our exposition follows Sandefur (1990), and additional details and examples can be found there and in the Bibliographic Notes in Section 3.7.

We shall motivate our discussion of univariate nonlinear dynamical systems with the basic *logistic equation* or *logistic dynamical system*,

$$\alpha_t = (1 + \theta_1)\alpha_{t-1} - \frac{\theta_1}{\theta_0}\alpha_{t-1}^2, \quad t = 1, 2, \dots, \quad (3.49)$$

which is an example of the more general system, $\alpha_t = \mathcal{M}(\alpha_{t-1})$. In population dynamics, (3.49) can be used to model density-dependent population growth. In that case, θ_1 is a growth parameter and θ_0 is the *carrying capacity* of the environment. The set of equilibrium values for the logistic equation is easily found to be $\{0, \theta_0\}$.

As with linear systems, we would like to characterize the system behavior near the equilibrium values. When $\mathcal{M}$ is a differentiable function (with respect to α_{t-1}), one can show that the equilibrium value a is attracting if $|\mathcal{M}'(a)| < 1$ and is repelling if $|\mathcal{M}'(a)| > 1$, where the superscript “ $(\cdot)$ ” denotes the first derivative with respect to a . Thus, for the logistic equation, $\mathcal{M}'(a) = 1 + \theta_1 - 2(\theta_1/\theta_0)a$. Then, if $\theta_1 > 0$ or $\theta_1 < -2$, the equilibrium value $a = 0$ is repelling; and if $-2 < \theta_1 < 0$, it is attracting. Similarly, $a = \theta_0$ is repelling if $\theta_1 > 2$ or $\theta_1 < 0$, and it is attracting if $0 < \theta_1 < 2$. Clearly, this suggests that the behavior of the system is quite different, depending on the values of the parameters θ_0 and θ_1 . In the context of a population model, a growth parameter θ_1 between 0 and 2 suggests a population that stabilizes at its carrying capacity, whereas a growth parameter value between 0 and -2 suggests a population that eventually becomes extinct (see Figure 3.4). It is then reasonable to ask what the system behavior for other values of these parameters is. In fact, this system can be quite complicated. However, before we give examples, we need a few more concepts.

One has to delve deeper to understand the dynamics at an equilibrium value a when the first derivative is inconclusive (i.e., $|\mathcal{M}'(a)| = 1$). For example, it can be shown that if $\mathcal{M}'(a) = 1$ and $\mathcal{M}''(a) \neq 0$ (where the superscript “ $(\cdot)$ ” denotes the second derivative with respect to a), then a is said to be *semistable*. For example, if $\mathcal{M}''(a) < 0$, we say a is *semistable from above*, which simply means that for values of α_{t-1} greater than a , the process is stable, but is unstable for values of α_{t-1} less than a . Similarly, if $\mathcal{M}''(a) > 0$, the process is *semistable from below* at a (i.e., stable for values of α_{t-1} less than a and unstable for values of α_{t-1} greater than a). If $\mathcal{M}''(a) = 0$, one then considers third-derivatives, if possible. The case where $\mathcal{M}'(a) = -1$ is a little more complicated, but one can in general examine the stability in this case as well, again based on consideration of the second (and third) derivatives of $\mathcal{M}$ at equilibrium value a (e.g., Sandefur, 1990).

As with the linear case, for a first-order nonlinear system, we say that two numbers a_1 and a_2 form a 2-cycle if when $\alpha_t = a_1$, then $\alpha_{t+1} = a_2$, $\alpha_{t+2} = a_1$,

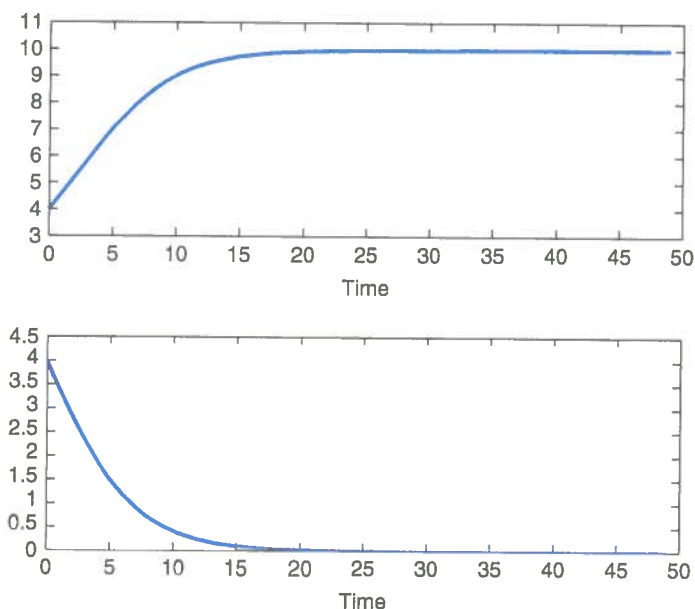


Figure 3.4 The logistic dynamical system (3.49) with starting value $\alpha_0 = 4$. **Top panel:** The case where $\theta_0 = 10$ and $\theta_1 = 0.25$. **Bottom panel:** The case where $\theta_0 = 10$ and $\theta_1 = -0.25$.

etc. In this case, the 2-cycle is stable or attracting if for intervals (γ_1, δ_1) and (γ_2, δ_2) around a_1 and a_2 , respectively, the initial condition $a_0 \in (\gamma_1, \delta_1)$ implies that as t increases, the limit of α_{2t} goes to a_1 , and the limit of α_{2t+1} goes to a_2 ; and, if $a_0 \in (\gamma_2, \delta_2)$, then as t increases, the limit of α_{2t} goes to a_2 , and the limit of α_{2t+1} goes to a_1 . In general, one can have a “ k -cycle,” consisting of the k values $a_1, \dots, a_k$, which appear in a repeating sequence as t increases. Such a cycle is stable if $|\mathcal{M}^{(l)}(a_1) \mathcal{M}^{(l)}(a_2) \dots \mathcal{M}^{(l)}(a_k)|$ is less than one but is unstable if this quantity is greater than one. As before, more complicated behavior occurs in situations where this quantity equals one (e.g., Sandefur, 1990).

3.2.3 Bifurcation

Because it is natural to think linearly, we often believe intuitively that small changes in inputs or parameter values lead to small changes in the behavior of the system. As we have already seen, small changes in parameter values can suggest quite different system behavior. Even more extreme cases exist where, for example, one value of a parameter suggests a completely different set of equilibrium points than a “nearby” value of the same parameter. Thus, the qualitative behavior of the system changes dramatically over relatively small changes in the parameter space. Values of the parameters where this happens are known as *bifurcation values*.

Bifurcations are very important in the study of dynamical systems. Consider a dynamical system that depends on parameter θ . Let n_θ be the number of equilibrium values for the system when the parameter θ takes a particular value. Now, if n_θ is not constant for θ in *any* interval $(\theta_0 - \epsilon, \theta_0 + \epsilon)$, then we say that θ_0 is a *bifurcation value* and the system is said to experience a *bifurcation* at θ_0 . One often plots so-called *bifurcation diagrams*, which are plots of all equilibrium values (on the y-axis) as a function of the parameter θ (plotted on the x-axis). Different types of bifurcations are defined based on the behavior indicated in these diagrams. As a simple example, consider the one-parameter logistic equation,

$$\alpha_t = \theta \alpha_{t-1} (1 - \alpha_{t-1}), \quad t = 1, 2, \dots, \quad (3.50)$$

for $\theta > 0$, which is a special case of (3.49). Then the equilibrium values are $a = 0$ and $a = (\theta - 1)/\theta$. The bifurcation diagram is shown in Figure 3.5. There is a *transcritical bifurcation* at $\theta = 1$ on this diagram (where the two lines intersect). One can show that for this system, $a = 0$ is attracting when $0 < \theta < 1$, semistable from above when $\theta = 1$, and repelling when $\theta > 1$. The critical value $a = (\theta - 1)/\theta$ is attracting when $1 < \theta \leq 3$, and it is repelling when $\theta < 1$ or $\theta > 3$.

Note that when $\theta > 3$, both equilibrium values are repelling. However, the story does not end there, and it is interesting to examine the behavior of the

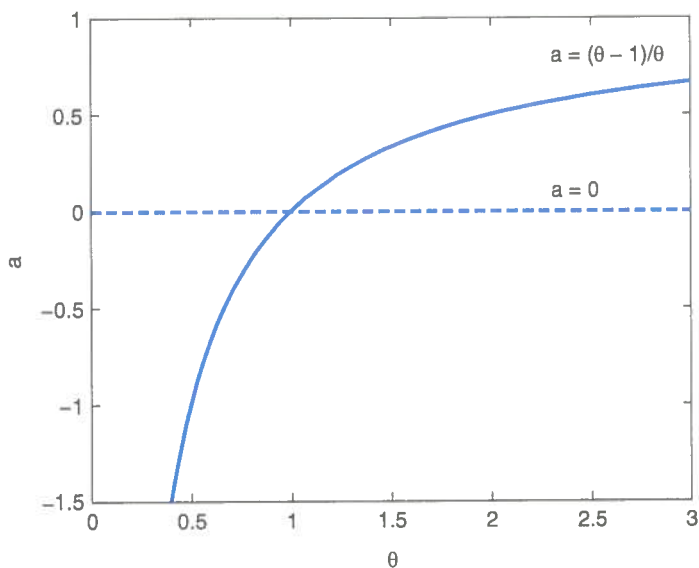


Figure 3.5 Bifurcation diagram for the one-parameter logistic equation (3.50); shown are equilibrium values as a function of θ . The solid blue line corresponds to the equilibrium values $a = (\theta - 1)/\theta$; the dashed blue line corresponds to the constant (as a function of θ) equilibrium value $a = 0$.

system in this case. One can show that for $3 < \theta < 1 + \sqrt{6}$, $\{\alpha_t\}$ behaves as an attracting 2-cycle, with attracting equilibrium values featured in the bifurcation diagram shown in Figure 3.6; the equilibrium values are given by

$$a = \sqrt{\theta + 1} \left(\frac{\sqrt{\theta + 1} \pm \sqrt{\theta - 3}}{2\theta} \right), \quad (3.51)$$

for $3 < \theta < 1 + \sqrt{6} \simeq 3.4495$. That is, there is a *period doubling* (or *pitch-fork*) bifurcation at $\theta = 3$. Now, if we increase θ so that $3.45 < \theta < 3.54$, there is an attracting 4-cycle. In fact, there is a sequence of numbers, $\theta_1 < \theta_2 < \dots < 3.57$, such that when $\theta_n < \theta < \theta_{n+1}$, the dynamical system $\{\alpha_t\}$ exhibits an attracting 2^n -cycle; that is, the system experiences repeated period doublings. For example, for $n = 12$ and $\theta_{12} < \theta < \theta_{13}$, $\{\alpha_t\}$ will exhibit an attracting 4096-cycle! Note that the behavior of $\{\alpha_t\}$ for $\theta > 3.57$ is even more complicated, with continued period-doubling as well as aperiodic behavior. Informally, for a given θ , the set of attracting fixed points (or regions) make up the *attractor*. We say that a dynamical system is *transitive* if, when the initial condition α_0 is “near” some point in the attractor, then for every point in the

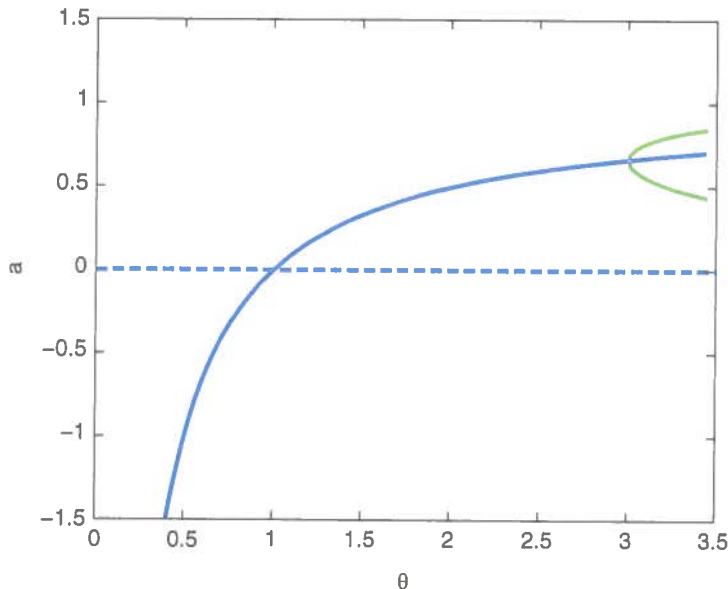


Figure 3.6 Bifurcation diagram for the one-parameter logistic equation (3.50), where now the horizontal axis is extended beyond $\theta = 3$. Shown are equilibrium values as a function of θ . The solid blue line corresponds to the equilibrium values $a = (\theta - 1)/\theta$; the dashed blue line corresponds to the constant (as a function of θ) equilibrium value $a = 0$. The solid green line corresponds to equilibrium values from an attracting 2-cycle, given by (3.51).

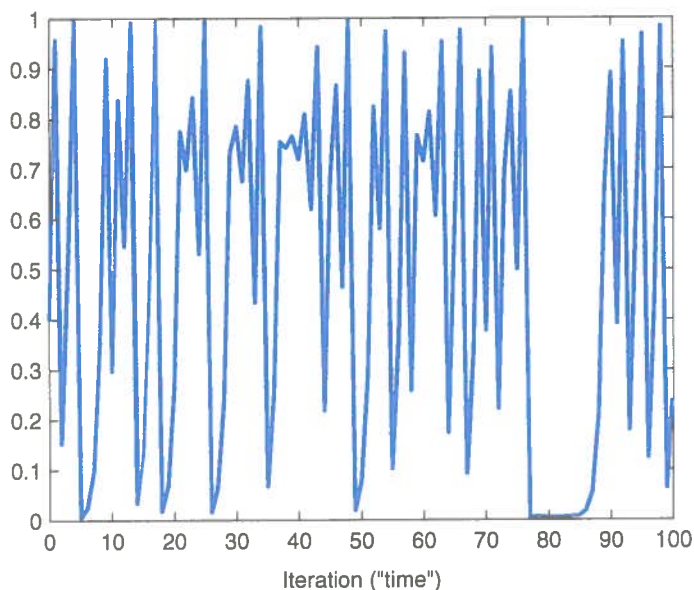


Figure 3.7 Iterations 0–100 of the one-parameter logistic equation (3.50) with $\theta = 4$ and $\alpha_0 = 0.4$.

attractor there is a sequence of values $\{\alpha_t\}$ that converges to that point. Eventually, when $\theta = 4$, the one-parameter logistic equation is said to be “chaotic.” Figure 3.7 shows iteration $t = 1$ to $t = 100$ for such a system. Note that the behavior is, to the eye at least, somewhat “random.”

3.2.4 Chaos

At this point, it is useful to ask what mathematical chaos is. Unfortunately, there is not a universally accepted definition. So far, we have seen several ways of characterizing complicated systems (e.g., bifurcations, attractors, transitivity). Yet, the colloquial expression of chaos is *sensitivity to initial conditions*. Informally, we say that a dynamical system exhibits sensitivity to initial conditions if, whenever there are two initial values $\alpha_0^{(1)}$ and $\alpha_0^{(2)}$ that are “close” together (i.e., $0 < |\alpha_0^{(1)} - \alpha_0^{(2)}| < \epsilon$, for some arbitrary ϵ), then the associated sequences $\alpha_t^{(1)}$ and $\alpha_t^{(2)}$ eventually get further apart (i.e., $|\alpha_t^{(1)} - \alpha_t^{(2)}| > \epsilon$, for $t > T$).

Consider the one-parameter logistic equation (3.50) with $\theta = 2$ and initial conditions close to the attracting equilibrium value $(\theta - 1)/\theta = 0.5$ (say, $\alpha_0^{(1)} = 0.510$ and $\alpha_0^{(2)} = 0.515$). In this case, there is no sensitivity to initial conditions, as the process is attracted to the equilibrium value, in either case, by the third iteration. Now let $\theta = 4$ with initial conditions, $\alpha_0^{(1)} = .510$ and $\alpha_0^{(2)} = .510001$. Then there is sensitivity to initial conditions as evidenced by the divergence of the realizations shown in Figure 3.8.

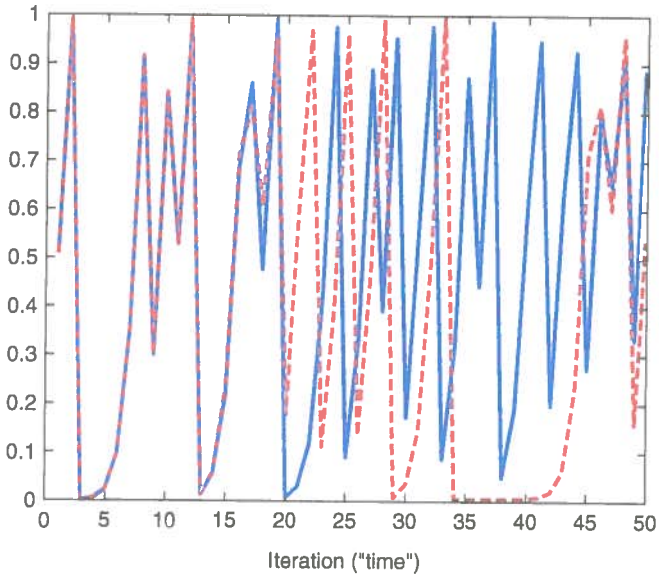


Figure 3.8 Two realizations showing iterations 0–50 of the one-parameter logistic equation (3.50) with $\theta = 4$ and two different starting values: Blue line, $\alpha_0 = 0.51$; red line, $\alpha_0 = 0.510001$.

As another example, consider the simple *linear* system, $\alpha_t = 2.5\alpha_{t-1}$. Clearly, this system shows sensitivity to initial conditions, since $|\alpha_t^{(1)} - \alpha_t^{(2)}| \rightarrow \infty$ for any $\alpha_0^{(1)} \neq \alpha_0^{(2)}$. But, this system does not have an attractor, which suggests that sensitivity to initial conditions is not sufficient to define the notion of chaos. One possible definition of chaos requires that the dynamical system:

- is transitive on its attractor,
- exhibits sensitive dependence on initial conditions,
- has repeating cycles that are “close” to the attractor.

The logistic equation (3.50) with $3.57 < \theta \leq 4$ exhibits chaos by this definition.

Consider a simple two-dimensional nonlinear system attributed to Hénon (1976):

$$\alpha_t = 1 + \beta_{t-1} - \theta_1 \alpha_{t-1}^2, \quad (3.52)$$

$$\beta_t = \theta_2 \alpha_{t-1}; \quad t = 1, 2, \dots, \quad (3.53)$$

for fixed parameters θ_1 and θ_2 . As in Berliner (1992), we let $\theta_1 = 1.4$, $\theta_2 = 0.3$, $\alpha_0 = 0.5$, and $\beta_0 = 0.4$. Figure 3.9 shows a simulation of this system. In addition, the scatter plot of realizations from this system, shown in Figure 3.10, illustrates the *attractor* (i.e., as time increases, the response of the system is

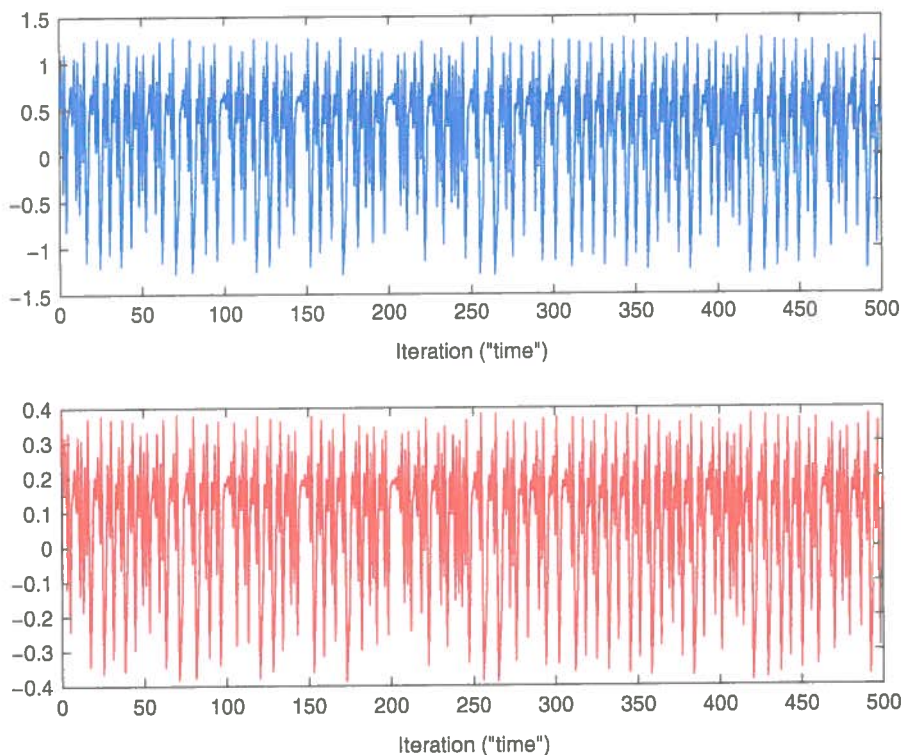


Figure 3.9 The first 500 iterations of the two-dimensional Hénon system (3.52) and (3.53) with $\theta_1 = 1.4$, $\theta_2 = 0.3$, and starting values $\alpha_0 = 0.5$, $\beta_0 = 0.4$. **Top panel:** $\{\alpha_t\}$. **Bottom panel:** $\{\beta_t\}$.

found in patterned geometrical regions of the state space). Clearly, this attractor has a complex geometrical structure and is, in fact, of fractal dimension. Informally, if all volumes in the state space contract as time increases, then the dynamical system is dissipative (i.e., the system will eventually evolve on a reduced set of states given by the attractor). Attractors that arise from chaotic dissipative systems are known as *strange attractors*. In contrast to other attractors, length elements of the geometrical region associated with the attractor will increase for some directions in state space. Yet, if length elements in other directions collapse sufficiently fast, the volume still shrinks overall. As described in Berliner (1992), there are competing mathematical forces at work in this case.

Characterizing Chaos

As we have discussed, sensitivity to initial conditions is a critical component of a chaotic system. However, it is not always so easy to verify such sensitivity. A related idea is the notion of a *Lyapunov exponent* (LE), which can typically be estimated from data. Intuitively, the LE measures the state-space contraction

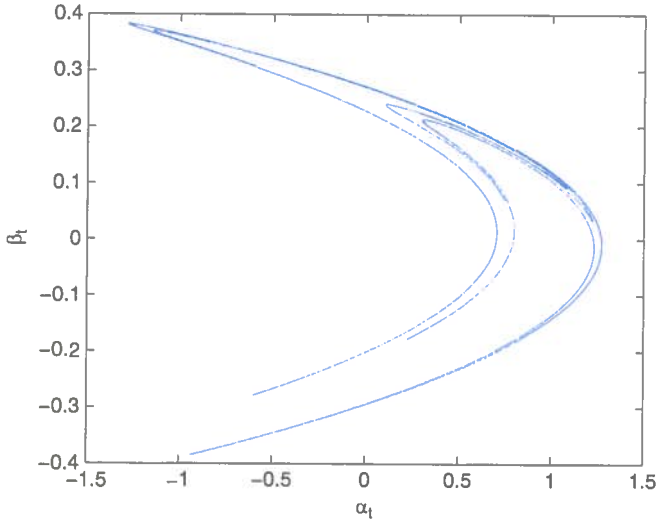


Figure 3.10 Scatterplot of 10,000 iterations of the Hénon system; the horizontal axis corresponds to α_t and the vertical axis corresponds to β_t .

or expansion. For illustration, assume the initial condition is represented by a hypersphere. For example, in two dimensions, (α_0, β_0) can be anywhere on the circle, $\alpha_0^2 + \beta_0^2 = c > 0$. As the system state evolves, the hypersphere transitions into a hyperellipsoid. The way in which the contraction or growth of each dimension (axis) of the ellipsoid relates to the initial condition can be characterized by a single coefficient.

Consider the LE for a scalar process: $\alpha_t = \mathcal{M}(\alpha_{t-1})$, where $\mathcal{M}$ is continuously differentiable with respect to α_{t-1} . Given two initial conditions, α_0 and α_0^* , the separation at time t of the associated states is given by

$$|\alpha_t^* - \alpha_t| = |\mathcal{M}'(\alpha_0^*) - \mathcal{M}'(\alpha_0)|, \quad (3.54)$$

where $\mathcal{M}'(\alpha) \equiv \mathcal{M}(\mathcal{M}(\dots \mathcal{M}(\alpha)))$. Now, expanding $\mathcal{M}'(\alpha_0^*) - \mathcal{M}'(\alpha_0)$ in a Taylor series around $\alpha_0^* = \alpha_0$, one can show that

$$|\alpha_t^* - \alpha_t| \simeq [\mathcal{M}^{(t)}(\alpha_0) \mathcal{M}^{(t)}(\alpha_1) \dots \mathcal{M}^{(t)}(\alpha_{t-1})] |\alpha_0^* - \alpha_0|, \quad (3.55)$$

where $\alpha_t = \mathcal{M}^t(\alpha_0)$. So, by taking appropriate limits, we obtain

$$\lim_{t \rightarrow \infty} |\alpha_t^* - \alpha_t| = e^{t\lambda(\alpha_0)} |\alpha_0^* - \alpha_0|, \quad (3.56)$$

where

$$\lambda(\alpha_0) \equiv \lim_{t \rightarrow \infty} \frac{1}{t} \sum_{k=0}^{t-1} \ln |\mathcal{M}^{(t)}(\alpha_k)| \quad (3.57)$$

is the *Lyapunov exponent*. Thus, one can think of the LE as measuring, on average, the asymptotic (exponential) rate of divergence of orbits with “close” initial conditions. In general, for identifying sensitivity to initial conditions, one looks for $\lambda(\alpha_0) > 0$. In this case, we say $1/\lambda(\alpha_0)$, sometimes known as the *Lyapunov time*, gives an average limit of predictability. However, we note that there can certainly be false indications of chaos in this context. For example, unbounded (periodic) orbits show positive LEs. In addition, it can be the case that orbits diverge, but not exponentially.

We can extend the notion of an LE to higher dimensions, as discussed in the Bibliographic Notes in Section 3.7. Also, we do not discuss the estimation of LEs here. We note that in general we cannot obtain LEs for nonlinear systems exactly, but there are numerical and statistical approaches for their estimation; again, see the Bibliographic Notes in Section 3.7.

3.2.5 State-Space Reconstruction

Consider a set of equations describing a multivariate dynamical system for a state process:

$$\alpha_t = \mathcal{M}(\alpha_{t-1}; \theta), \quad (3.58)$$

where $\mathcal{M}$ is a vector-valued function and θ is a vector of known parameters. If α_{t-1} completely specifies the state of the system, then exact knowledge of α_0 , $\mathcal{M}$, and θ allows the future state to be computed exactly. As we have already seen, lack of perfect knowledge of α_0 or θ can have tremendously important consequences on one’s ability to predict the future. What about knowledge of the state itself?

In reality, one would seldom have access to all of the state variables that make up the true state α_t . As an extreme case, say we could only measure one component of the system, say Z_t (a scalar), but we measure it perfectly! This leads to a measurement equation,

$$Z_t = h(\alpha_t), \quad (3.59)$$

where h is a measurement function. An impressive result from the theory of dynamical systems suggests that for systems of the form (3.58), we may not need to observe the entire state vector to represent the dynamics, but that noise-free observations of only one component (or some lower-dimensional subset) of the state is enough to reconstruct the system. Specifically, we define a *delay coordinate function* H that constructs an M -dimensional delay vector, $\mathbf{Y}_t \in \mathbb{R}^m$, from M observations separated by *delay time* τ_d :

$$\begin{aligned} \mathbf{Y}_t &\equiv H(\alpha_t, \alpha_{t-\tau_d}, \dots, \alpha_{t-(m-1)\tau_d}) \\ &\equiv (h(\alpha_t), h(\alpha_{t-\tau_d}), \dots, h(\alpha_{t-(m-1)\tau_d}))' \\ &= (Z_t, Z_{t-\tau_d}, \dots, Z_{t-(m-1)\tau_d})'. \end{aligned} \quad (3.60)$$

Takens (1981) provides the conditions under which such a delay vector can represent the dynamical system. Informally, this suggests that we do not have to measure all of the state variables of a system (which is a good thing, since we typically cannot!), but we can represent the system equivalently by time-lagged scalar noise-free observations of some subset of the system states. Unfortunately, in the real world this is not always practical, because one does not know the form of H , nor is it possible to have truly noise-free observations. So, the conditions of Takens' theorem may not be met in practice. Nevertheless, these ideas can provide useful motivation for building stochastic state-space models and, in some cases, can be used effectively to model real-world time series (e.g., Sugihara and May, 1990). We shall use them in Section 6.6.1 and Chapter 7 when we consider dimension reduction in spatio-temporal statistical models.

3.2.6 The Need for Statistical Models for Dynamical Processes

As we have seen, uncertainty breeds “chaos”; and, given that we always have uncertainty in observations, processes, and parameters, how should we best model real-world processes? Berliner (1992, Section 2.2) provides a nice discussion of randomness and chaos. In particular, he describes how “uncertainty, especially in the presence of complexity, naturally leads to the use of random or probabilistic methods.” In essence, the presence of uncertainties in our knowledge of real-world systems, at least at some scales of variability, suggests the need to characterize that uncertainty via probability models. Even if it were possible to know exactly the dynamics of a real-world system, we would still be faced with uncertainties related to our observations of the system, which, of course, lead to issues related to sensitivity to initial conditions. Since it is often not possible to distinguish between such a chaotic effect and true randomness (e.g., a Bernoulli coin toss), one might benefit from modeling a chaotic system as a random system. In fact, a chaotic system such as the logistic map considered in (3.50) can formally be linked to a zero-one process such as the Bernoulli process, thereby creating a zero-one generating mechanism from a deterministic dynamical system that is as rich (and appears as “random”) as the Bernoulli process [see Berliner (1992)]. Furthermore, Berliner (1992) discusses the link between *ergodic theory* and chaos.

Although we shall use *notions* from deterministic dynamical-system theory in our discussion of spatio-temporal statistical analysis, modeling of real-world problems involves uncertainty that is best handled in a probability-based framework. Thus, in the rest of this chapter, we consider *stochastic dynamical systems*, which are simply (random) time series.

3.3 TIME SERIES PRELIMINARIES

From the introduction to this chapter, recall that time series models can be viewed as process models for temporal data. Before we discuss them, we must first define some basic functions of a discrete-time sequence of real-valued

random variables $\{Y_t : t \in D_t\}$. Henceforth, we assume that $D_t = \{0, 1, \dots\}$, and we refer to $\{Y_t : t = 0, 1, \dots\}$ as a *time series*.

The *mean function* is defined as

$$\mu_t \equiv E(Y_t), \quad t \in D_t, \quad (3.61)$$

which is simply the mean of the process relative to the underlying probability space. Then $\{\mu_t : t = 0, 1, \dots\}$ is deterministic and it is quite possible that it satisfies a dynamical system of the sort discussed in Section 3.2. Indeed, this will be a key component of certain hierarchical spatio-temporal dynamical models discussed in Chapter 7. However, more typically, we choose simpler models for μ_t and put much of the uncertainty in the remaining stochastic component, $\{Y_t - \mu_t : t = 0, 1, \dots\}$. This uncertainty can be described through the *autocovariance function* or *covariogram*,

$$C_Y(t, r) = \text{cov}(Y_t, Y_r) = E\{(Y_t - \mu_t)(Y_r - \mu_r)\}, \quad t, r \in D_t, \quad (3.62)$$

which summarizes how the process co-varies across different time lags, after accounting for the mean function. Notice that $C_Y(t, r) = C_Y(r, t)$ and, furthermore, that the variance is a special case of the autocovariance in which $C_Y(t, t) = \text{var}(Y_t) = \sigma_t^2$. After normalization, one can obtain the *autocorrelation function*, a function that measures linear statistical dependence between different members of the time series:

$$\rho_Y(t, r) \equiv \frac{C_Y(t, r)}{\sqrt{C_Y(t, t)C_Y(r, r)}}, \quad t, r \in D_t, \quad (3.63)$$

where $\rho_Y(t, r) \in [-1, 1]$. A sufficient condition for (3.62) and (3.63) to exist is $\sigma_t^2 < \infty$, for all D_t .

In Section 2.1, we discussed information content of a dataset and pointed out that, without further assumptions, time series data could be viewed as a sample of size one! One type of assumption commonly made for time series models is that of *stationarity*. There are different forms of stationarity; and to avoid repeating the definitions given in Section 2.2, for d -dimensional space, we simply summarize them here for the special case of one dimension (which here is time). Note that there is further discussion of stationarity in the spatio-temporal context in Section 6.1.2.

Strong stationarity of $\{Y_t\}$ is an assumption that says that any finite collection, $\{Y_{t_1}, \dots, Y_{t_m}\}$, of random variables from the time series has exactly the same joint distribution as $\{Y_{t_1+\tau}, \dots, Y_{t_m+\tau}\}$, for any $\tau \in \{0, \pm 1, \pm 2, \dots\}$. Note that the same definition applies to continuous-time temporal processes $\{Y(t) : t \in D_t\}$, but where the time lag is $\tau \in (-\infty, \infty)$.

Now, it is possible that a probability distribution may not have any moments! For example, the Cauchy distribution has no mean, variance, third moment, etc. Weak stationarity of $\{Y_t\}$ is an assumption that requires initially the *existence* of the second moment (then the first moment automatically exists); that is, assume

$\text{var}(Y_t) \equiv \sigma_t^2 < \infty$, for all $t \in D_t$. Time series whose variances are finite are said to be *second-order stationary* (equivalently, *weakly stationary*) if

- (i) $E(Y_t) \equiv \mu$, for all $t \in D_t$.
- (ii) $\text{cov}(Y_t, Y_r) \equiv C_Y(t - r)$, for all $t, r \in D_t$.

Second-order stationarity is an assumption that says that any pair, $\{Y_t, Y_r\}$ has exactly the same first and second moments (including the cross-moment which defines the autocovariance function) as the pair $\{Y_{t+\tau}, Y_{r+\tau}\}$, for any $\tau \in \{0, \pm 1, \dots\}$. For processes with finite variance, strong stationarity implies second-order stationarity, but not vice versa.

Even if μ_t given by (3.61) is not constant, we can define the *autocovariance function* to be *stationary* if the covariance function given by (3.62) is equal to $C_Y(t - r)$, for all $t, r \in D_t$. Note that the same definitions of second-order stationarity apply to continuous-time temporal processes $\{Y(t)\}$. Finally, a discussion of ergodicity and its relation to strong stationarity is given in Cressie (1993, pp. 53–58).

In addition to characterizing the second-order dependence structure of a time series in terms of a stationary autocovariance (or autocorrelation) function, an equivalent characterization is in terms of a Fourier spectral representation. This formal equivalence is given by the *spectral representation theorem* (e.g., see Shumway and Stoffer, 2006, Appendix C).

In the rest of this chapter, we write the stationary covariance function as $C(\tau)$, rather than $C_Y(\tau)$. Given a second-order stationary time series $\{Y_t\}$ with autocovariance function $C(\tau) \equiv E\{(Y_{t+\tau} - \mu)(Y_t - \mu)\}$, a *spectral distribution function* $F(\omega)$ exists that is monotone, nondecreasing, and bounded for $-1/2 \leq \omega \leq 1/2$, with $F(-1/2) = 0$, $F(1/2) = C(0)$, such that

$$C(\tau) = \int_{-1/2}^{1/2} \exp\{2\pi i \omega \tau\} dF(\omega), \quad \tau = 0, \pm 1, \pm 2, \dots \quad (3.64)$$

For a stationary process with autocovariance function satisfying, $\sum_{\tau=-\infty}^{\infty} |C(\tau)| < \infty$ (absolute summability), the spectral distribution function is absolutely continuous with $dF(\omega) = f(\omega) d\omega$, and hence

$$C(\tau) = \int_{-1/2}^{1/2} \exp\{2\pi i \omega \tau\} f(\omega) d\omega, \quad \tau = 0, \pm 1, \pm 2, \dots \quad (3.65)$$

Further, the *spectral density function* $f(\omega)$ can be represented as

$$f(\omega) = \sum_{\tau=-\infty}^{\infty} C(\tau) \exp\{-2\pi i \omega \tau\}, \quad -1/2 \leq \omega \leq 1/2. \quad (3.66)$$

Note that because time is discrete here, we can interpret the spectral density function (3.66) as the Fourier transform of the autocovariance function and, the autocovariance function (3.65) as the inverse Fourier transform of the spectral

density function. Furthermore, the spectral density function is nonnegative over all ω ; it is an even function, $f(\omega) = f(-\omega)$; and it has a period of one, $f(\omega) = f(\omega + 1)$.

In the context of a *multivariate* time series $\{Y_t\}$, we define $\{Y_t^{(i)}\}$, to be the i th time series at time t . Then, in general, we define

$$C_{ij}(t, r) \equiv \text{cov}(Y_t^{(i)}, Y_r^{(j)}), \quad t, r \in D_t, \quad (3.67)$$

to be the *cross-covariance function* between the i th time series at time t and the j th time series at time r . A second-order stationarity assumption on the multivariate time series $\{Y_t\}$ requires that the respective mean functions do not vary with time and that the cross-covariance functions only depend on the lag between times t and r . That is,

$$C_{ij}(\tau) = \text{cov}(Y_t^{(i)}, Y_{t+\tau}^{(j)}), \quad \tau = 0, \pm 1, \pm 2, \dots \quad (3.68)$$

Note that $C_{ij}(-\tau) = C_{ji}(\tau)$. Furthermore, the *cross-correlation function* is given by

$$\rho_{ij}(\tau) = \frac{C_{ij}(\tau)}{\sqrt{C_{ii}(0)C_{jj}(0)}}, \quad \tau = 0, \pm 1, \pm 2, \dots \quad (3.69)$$

Empirical Mean and Empirical Covariance Function

Assume that we have a finite sample from a temporal process $\{Y_t\}$, denoted by $Y_1, \dots, Y_T$, and that there is only this one realization available to summarize the time series. The formulas that follow apply whether $\{Y_t\}$ is deterministic or stochastic, but they have an interpretation as estimators when the time series is second-order stationary.

The *empirical mean* (also called the *sample mean*) of the time series is

$$\hat{\mu} \equiv \frac{1}{T} \sum_{t=1}^T Y_t. \quad (3.70)$$

The empirical (or *sample*) *autocovariance function* is

$$\hat{C}(\tau) \equiv \frac{1}{T} \sum_{t=1}^{T-\tau} (Y_{t+\tau} - \hat{\mu})(Y_t - \hat{\mu}), \quad \tau = 0, 1, \dots, T-1. \quad (3.71)$$

Note that $\hat{C}(-\tau) = \hat{C}(\tau)$ and $\hat{C}(0)$ is the *empirical* (or *sample*) *variance*. One could divide the sum in (3.71) by $(T - \tau)$ rather than T , and in fact we do this in Chapter 5. Dividing by T in (3.71) guarantees that the empirical autocovariance function is nonnegative definite, which guarantees that the empirical variances of linear combinations of the series, $\{Y_t\}$, will be nonnegative (e.g., Shumway and Stoffer, 2006, Section 1.6). However, (3.71) is clearly biased; and to mitigate this effect for exploratory spatio-temporal data

analysis, we divide by $T - \tau$ in Chapter 5. Similarly, the *empirical* (or *sample*) *autocorrelation function* is

$$\hat{\rho}(\tau) \equiv \frac{\hat{C}(\tau)}{\hat{C}(0)}, \quad \tau = 0, 1, \dots, T - 1. \quad (3.72)$$

In the multivariate context, assume that we have observations from a process $\{\mathbf{Y}_t\}$; recall that observations on the i th time series are denoted by $Y_1^{(i)}, \dots, Y_T^{(i)}$. The *empirical* (or *sample*) *cross-covariance function* is

$$\hat{C}_{ij}(\tau) \equiv \frac{1}{T} \sum_{t=1}^{T-\tau} (Y_{t+\tau}^{(i)} - \hat{\mu}_i)(Y_t^{(j)} - \hat{\mu}_j), \quad \tau = 0, 1, \dots, T - 1, \quad (3.73)$$

where $\hat{\mu}_i$ is the empirical mean (3.70) for the i th time series and likewise for $\hat{\mu}_j$. Again, the divisor T could be replaced by $T - \tau$ (Chapter 5). We note that $\hat{C}_{ij}(-\tau) = \hat{C}_{ji}(\tau)$. Finally, the empirical cross-correlation function is obtained from (3.69) after substituting the respective empirical covariance and cross-covariance functions from (3.73).

When viewed as estimators, a discussion of the statistical properties of the empirical first and second moments presented here can be found in most advanced time series books. For more details, see the references in the Bibliographic Notes in Section 3.7.

3.4 BASIC TIME SERIES MODELS

In addition to the assumption of stationarity, probabilistic properties of time series are typically specified in order to simplify their characterization. The fact that the time series literature emphasizes such process-model-based descriptions is motivated by Science. This is in contrast to much of the literature in geostatistics (e.g., see Section 4.1), where, for historical reasons, more descriptive (e.g., covariance-based) and less explanatory models are used to represent the data. Geostatistics was originally used in the mining industry for prediction of the mineral content of ore bodies. While the geologist's knowledge is crucial for locating the ore body, geostatistical prediction has typically relied on (spatial) autocovariances, often in the form of variograms, estimated from ore samples. These ideas are taken up in Section 4.1.

Basic process models for time series are described briefly below. Here, we focus only on discrete-time processes, although in most cases analogous models exist for continuous time.

3.4.1 White-Noise Process

We define a *white-noise* process as a discrete-time random process $\{W_t : \dots, -1, 0, 1, \dots\}$ whose elements are mutually independent and have a

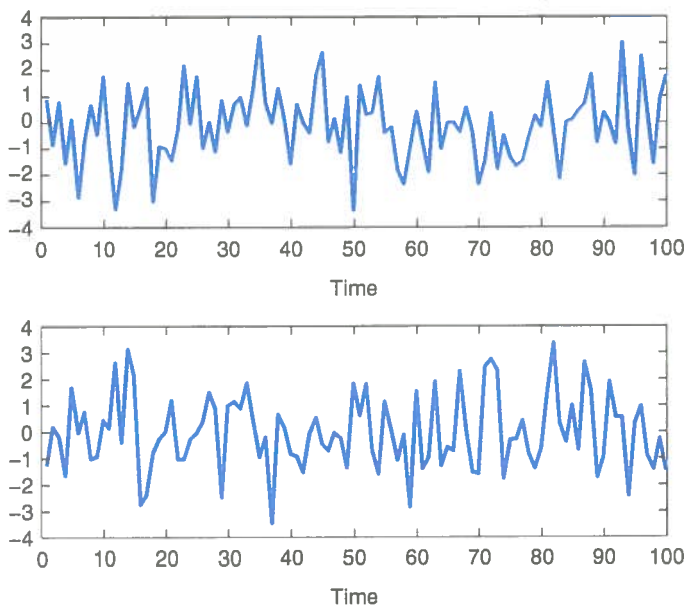


Figure 3.11 Two independent realizations of length 100 of a white-noise process, $W_t \sim iid\ Gau(0, 2)$.

common probability density function (pdf). Typically, its mean μ_w is assumed to be zero, and its autocovariance function is

$$C(\tau) = \begin{cases} \sigma_w^2, & \tau = 0, \\ 0, & \tau = \pm 1, \pm 2, \dots, \end{cases} \quad (3.74)$$

where $\sigma_w^2 > 0$ is the white-noise variance. From the definition of the spectral density function, (3.66), a white-noise process has, $f(\omega) = \sigma_w^2$, for $-1/2 < \omega < 1/2$. That is, the spectral density is equal for all frequencies, which is analogous to how white light is a mixture of colors with no preference in the color spectrum (hence the name “white noise”).

An example of a white-noise process is $W_t \sim iid\ Gau(0, \sigma_w^2)$, for all t . Figure 3.11 shows two (independent) realizations from the same white-noise process with variance $\sigma_w^2 = 2$. Although it is common that a white-noise process is Gaussian, it need not be.

3.4.2 Random-Walk Process

Some processes in the natural world behave in a manner that suggests their value at the current time is simply their value at the previous time plus some noise. More formally, a time series $\{Y_t\}$ is said to be a *random walk* if

$$Y_t = Y_{t-1} + W_t, \quad t = 1, 2, \dots, \quad (3.75)$$

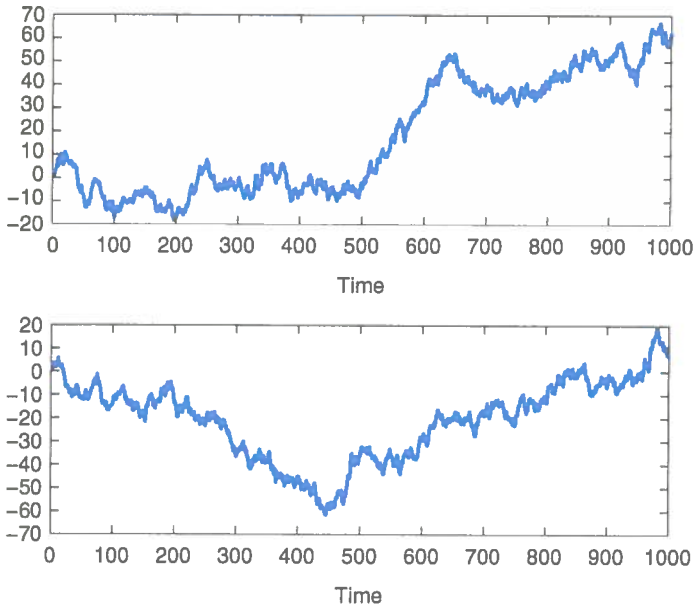


Figure 3.12 Two independent realizations of length 1000 of a random walk process with $Y_0 \equiv 0$, and $W_t \sim iid\ Gau(0, 2)$.

where $\{W_t\}$ is a white-noise process with mean μ_w and variance σ_w^2 . One can show that in this case, $E(Y_t) = E(Y_0) + t\mu_w$ and $\text{var}(Y_t) = t\sigma_w^2$. That is, both the mean and variance are functions of time and, thus, the process is nonstationary. Furthermore, $\text{cov}(Y_t, Y_r) = \min(t, r)\sigma_w^2$.

Figure 3.12 shows two independent realizations from the same random-walk process with $W_t \sim iid\ Gau(0, 2)$. Note that such processes exhibit behavior that is “locally smooth” but quite obviously random.

3.4.3 Autoregressive Process

Often, observations of environmental, ecological, biological, and physical processes depend on one or more observations that immediately precede it. A time series that models such structure is the *autoregressive* (or AR) process. A time series $\{Y_t\}$ is said to be an *autoregressive process of order p* , $AR(p)$, if

$$Y_t = m_1 Y_{t-1} + m_2 Y_{t-2} + \cdots + m_p Y_{t-p} + W_t, \quad t = \dots, -2, -1, 0, 1, 2, \dots, \quad (3.76)$$

where $\{W_t\}$ is a white-noise process with mean zero and variance σ_w^2 , and where $\{m_i : i = 1, \dots, p\}$, are fixed but unknown parameters.

For example, consider the $AR(1)$ process,

$$Y_t = m_1 Y_{t-1} + W_t, \quad t = \dots, -2, -1, 0, 1, 2, \dots, \quad (3.77)$$

where $\{W_t\}$ is white-noise process with mean zero and variance σ_w^2 , and we write $\alpha = \alpha_1$ for notational simplicity. By back-substitution, we obtain

$$\begin{aligned} Y_t &= W_t + \alpha W_{t-1} + \alpha^2 W_{t-2} + \alpha^3 W_{t-3} + \cdots \\ &= \sum_{k=0}^{\infty} \alpha^k W_{t-k}, \end{aligned} \quad (3.78)$$

where it is assumed that $|\alpha| < 1$. That is, the AR(1) process can be written as an infinite series of white-noise random variables (i.e., an infinite-order *moving-average process*, as will be discussed in Section 3.4.4). Since $E(W_t) = 0$ and $\text{var}(W_t) = \sigma_w^2$, it follows that $E(Y_t) = 0$ and

$$\text{var}(Y_t) = \sigma_w^2(1 + \alpha^2 + \alpha^4 + \cdots) = \frac{\sigma_w^2}{1 - \alpha^2}, \quad (3.79)$$

which does not depend on t . If the condition $|\alpha| < 1$ is not met, the process is explosive (i.e., the variance of the time series increases exponentially with time). Furthermore, when $\alpha = 1$, the process is a random walk. In the explosive case where $|\alpha| > 1$, one can show that $Y_t = -\sum_{k=1}^{\infty} \alpha^{-k} W_{t+k}$, which is stationary (e.g., Shumway and Stoffer, 2006, Section 3.2). However, such a model requires knowledge of the future and is thus not helpful. We say that a temporal process that does not depend on the future is *causal*.

It is also straightforward to show that for the AR(1) process with $|\alpha| < 1$, the autocorrelation function is given by

$$\rho(\tau) = \alpha^{|\tau|}, \quad \tau = 0, \pm 1, \pm 2, \dots \quad (3.80)$$

Thus, for the AR(1) process, the autocorrelation (i.e., temporal dependence) decreases in magnitude as the time lag τ increases. From (3.66), the spectral density for the AR(1) process is given by

$$f(\omega) = \frac{\sigma_w^2}{1 - 2\alpha \cos(2\pi\omega) + \alpha^2}, \quad -1/2 < \omega < 1/2. \quad (3.81)$$

When the AR(1) parameter α is greater than 0, the spectral density decreases as $|\omega|$ increases. This is sometimes known as a *red-noise* spectrum, again analogous to the color spectrum. When α is less than 0, the spectrum increases as $|\omega|$ increases, which is sometimes known as a *blue-noise* spectrum. See Figure 3.13 for an illustration of these two cases.

Figure 3.14 shows three realizations of an AR(1) process with different values of α ; positive dependence (top panel), negative dependence (middle panel), and explosive growth (bottom panel) are featured.

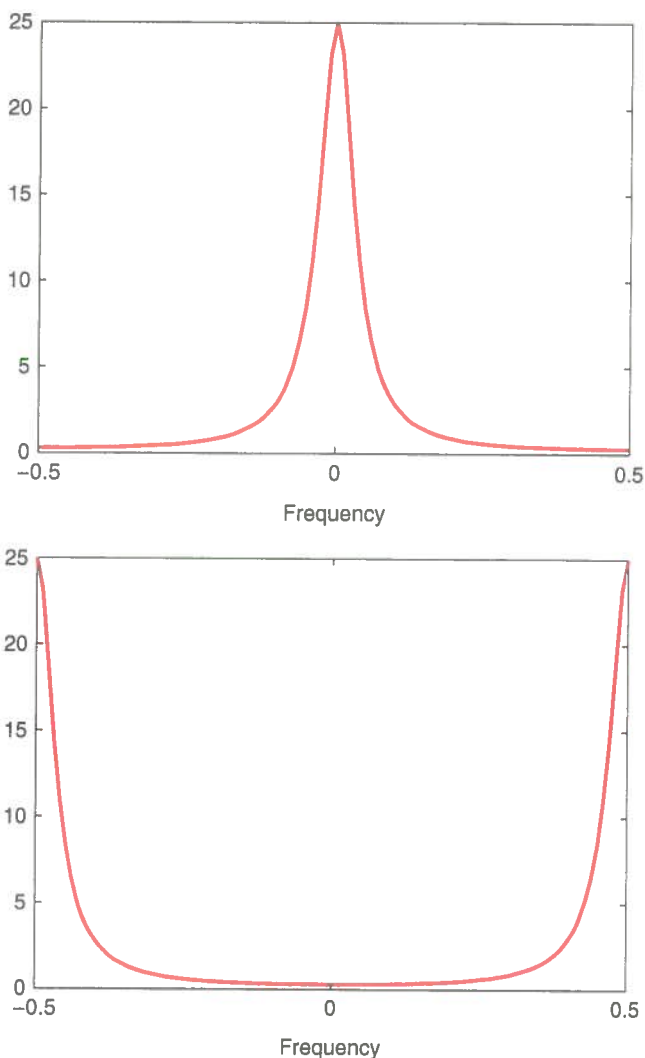


Figure 3.13 AR(1) spectral densities given by (3.81). **Top panel:** $\alpha = 0.8$. **Bottom panel:** $\alpha = -0.8$.

Autocovariance and autocorrelation functions for higher-order causal AR processes can be obtained fairly easily. For example, the AR(2) process,

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + W_t, \quad t = 2, 3, \dots, \quad (3.82)$$

has an autocorrelation function that satisfies,

$$\rho(\tau) = \alpha_1 \rho(\tau - 1) + \alpha_2 \rho(\tau - 2), \quad \tau = 1, 2, \dots, \quad (3.83)$$

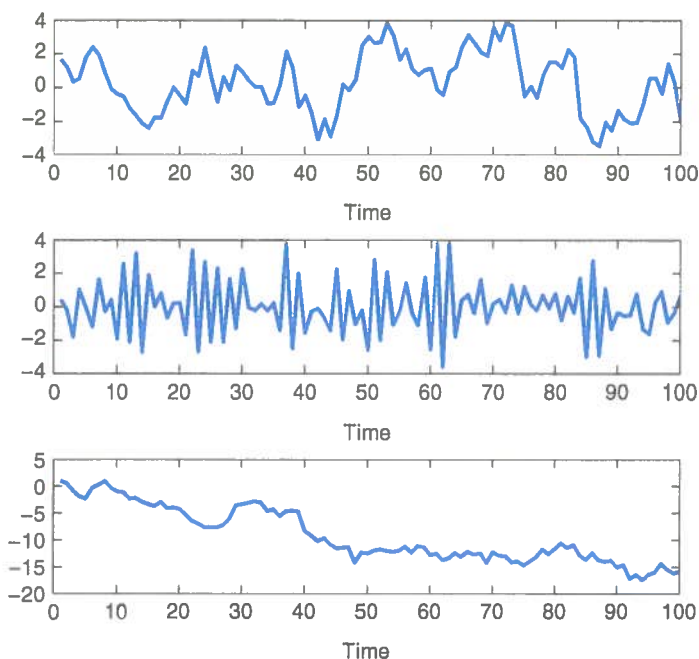


Figure 3.14 AR(1) realizations, defined by (3.77), of length 100. **Top panel:** $m_1 = 0.8$ (analogous to the spectral density in the top panel of Figure 3.13). **Middle panel:** $m_1 = -0.8$ (analogous to the spectral density in the bottom panel of Figure 3.13). **Bottom panel:** $m_1 = 1.01$ (explosive growth). In all cases, $\sigma_w^2 = 1$.

which can be obtained by multiplying both sides of (3.82) by $Y_{t-\tau}$ and taking expectations. One can study the behavior of this function by noting that (3.83) is a second-order deterministic dynamical system (see Section 3.2.1). In particular, given initial conditions, $\rho(0) = 1$ and $\rho(-1) = \alpha_1/(1 - \alpha_2)$, one can show that the autocorrelation function will exhibit exponential decay with increasing lag τ , but may show oscillatory behavior as well (see Section 3.2.1). This behavior is also evident upon examination of the AR(2)'s spectral density function:

$$f(\omega) = \frac{\sigma_w^2}{1 + m_1^2 + m_2^2 - 2m_1(1 - m_2)\cos(2\pi\omega) - 2m_2\cos(4\pi\omega)}, \quad (3.84)$$

for $-1/2 < \omega < 1/2$. Depending on the values of m_1 and m_2 , the spectrum can exhibit increased power at the high or low frequencies, similar to the AR(1) process. In addition, as shown in Figure 3.15, the AR(2) spectrum can also accommodate a spectral peak (associated with oscillatory behavior of the process), which suggests that the AR(2) process is quite flexible in its ability to model real-world phenomena.

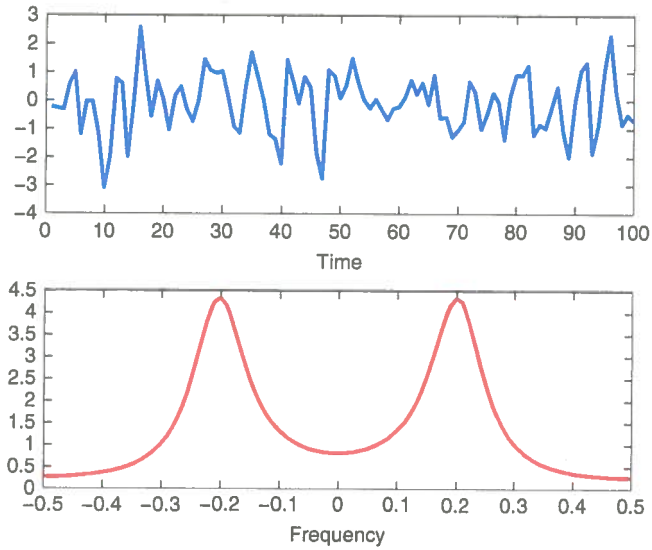


Figure 3.15 Top panel: A realization from an AR(2) process, defined by (3.82), of length 100; $m_1 = .4$, $m_2 = -.5$, and $\sigma_w^2 = 1$. Bottom panel: Theoretical spectral density (3.84) for the AR(2) process shown in the top panel.

Parameter Estimation

Given a finite sample $Y_1, \dots, Y_T$ from (3.76), we could use the empirical covariance function to estimate the parameters of the AR(p) model, namely $m_1, \dots, m_p$, and σ_w^2 . Multiply each side of (3.76) by Y_{t-j} , for $j = 0, \dots, p$, and take expectations. This results in the set of equations

$$C(0) = m_1 C(1) + \dots + m_p C(p) + \sigma_w^2 \quad (3.85)$$

$$C(k) = m_1 C(k-1) + \dots + m_p C(k-p), \quad k = 1, \dots, p, \quad (3.86)$$

which are known as the *Yule–Walker equations*. In matrix form, we can write these as

$$\mathbf{C}\mathbf{m} = \mathbf{c} \quad \text{and} \quad \sigma_w^2 = C(0) - \mathbf{m}'\mathbf{c}, \quad (3.87)$$

where $\mathbf{C}$ is a $p \times p$ matrix with (i, j) th element $C(i-j)$, and $\mathbf{m} \equiv (m_1, \dots, m_p)'$ and $\mathbf{c} \equiv (C(1), \dots, C(p))'$ are p -dimensional vectors. Method-of-moment estimators for $\mathbf{m}$ are obtained by replacing the autocovariance function $C(\cdot)$ with its empirical version $\hat{C}(\cdot)$ given by (3.71):

$$\hat{\mathbf{m}} \equiv \hat{\mathbf{C}}^{-1}\hat{\mathbf{c}} \quad (3.88)$$

and

$$\hat{\sigma}_w^2 = \hat{C}(0) - \hat{\mathbf{c}}'\hat{\mathbf{C}}^{-1}\hat{\mathbf{c}}. \quad (3.89)$$

Note that the Yule–Walker-based method-of-moments estimators (3.88) and (3.89) can be computed through the Durbin–Levinson algorithm, which does not require inversion of $\mathbf{C}$ (e.g., see Shumway and Stoffer, 2006, Section 3.5).

One can also obtain estimators of AR parameters via ordinary-least-squares and maximum-likelihood procedures. For the causal autoregressive process conditioned on the first p observations, all of these estimators give the same asymptotically normal limiting distribution [e.g., see Brockwell and Davis (1991, Chapter 8) and Fuller (1996, Chapter 8)]. Such results allow one to obtain large-sample confidence regions for model parameters. Note that in practice, a noisy version of the time series $Z_1, \dots, Z_T$ is observed, where typically $Z_t = Y_t + \varepsilon_t$, with $\{\varepsilon_t\}$ an independent white-noise process, mean 0 and variance σ_ε^2 (see Section 3.6). This requires only slight modification to (3.88) and (3.89), provided that σ_ε^2 is known or estimated from calibration data.

3.4.4 Moving-Average Process

The time series $\{Y_t\}$ is said to be a *moving-average (or MA) process of order q* , or $\text{MA}(q)$, if

$$Y_t = W_t + \beta_1 W_{t-1} + \dots + \beta_q W_{t-q}, \quad (3.90)$$

where $\beta_1, \beta_2, \dots, \beta_q$ are fixed but unknown parameters, and $\{W_t\}$ is a white-noise process with zero mean and variance σ_w^2 . At first glance, this may appear to be similar to a multiple-regression model. However, the key differences are that $\{W_t\}$ are random and, more importantly, they are unobservable!

It follows that for such a model, $E(Y_t) = 0$ and $\text{var}(Y_t) = \sigma_w^2 \sum_{i=0}^q \beta_i^2$, which does not depend on t . Furthermore,

$$\rho(\tau) = \begin{cases} 1, & \tau = 0, \\ \sum_{i=0}^{q-\tau} \beta_i \beta_{i+\tau} / \sum_{i=0}^q \beta_i^2, & |\tau| = 1, 2, \dots, q, \\ 0, & |\tau| > q, \end{cases} \quad (3.91)$$

where the process is standardized by setting $\beta_0 = 1$. One can then use (3.66) to obtain the spectral density. For example, the spectral density for the $\text{MA}(1)$ process, $Y_t = W_t + \beta W_{t-1}$, is given by

$$f(\omega) = \sigma_w^2 (1 + \beta^2 + 2\beta \cos(2\pi\omega)), \quad -1/2 < \omega < 1/2, \quad (3.92)$$

where we write the $\text{MA}(1)$ parameter $\beta = \beta_1$, for notational simplicity. The spectral density function is clearly dependent on the value of this parameter. When $\beta < 0$, more power is associated with the higher (in magnitude) frequencies; and when $\beta > 0$, more power is associated with the lower (in magnitude) frequencies (see Figure 3.16).

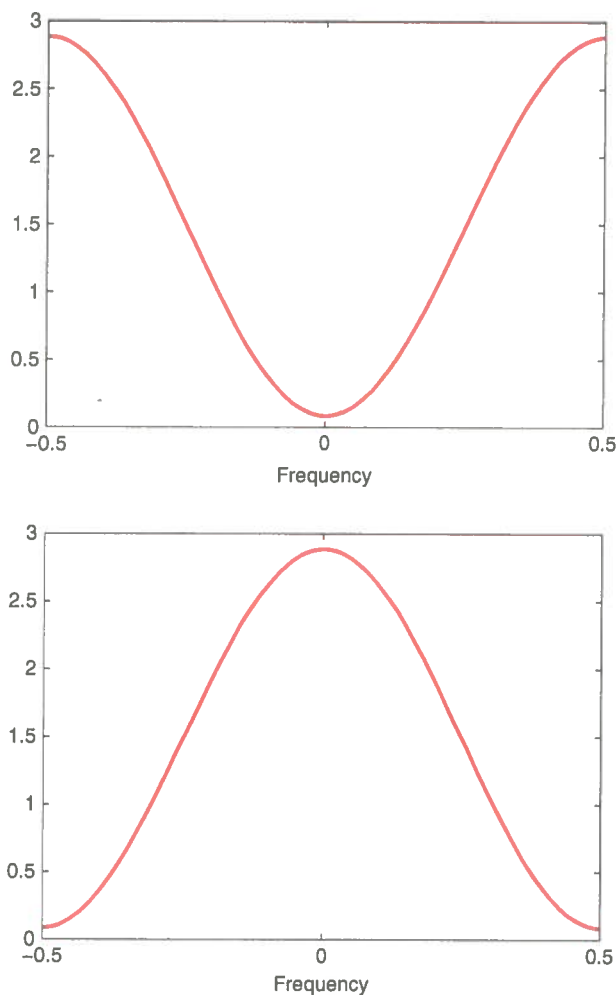


Figure 3.16 MA(1) spectral density given by (3.92). **Top panel:** $\beta = -0.7$. **Bottom panel:** $\beta = 0.7$.

An interesting feature of the MA process is illustrated by considering two different formulations of an MA(1) process (e.g., Chatfield, 1989):

$$Y_t = W_t + \beta W_{t-1}, \quad (3.93)$$

$$Y_t = W_t + \frac{1}{\beta} W_{t-1}. \quad (3.94)$$

From (3.91), it is clear that both models have exactly the same autocorrelation function. Now, from (3.93), we obtain the following representation by back-substitution:

$$W_t = Y_t - \beta Y_{t-1} + \beta^2 Y_{t-2} - \dots, \quad (3.95)$$

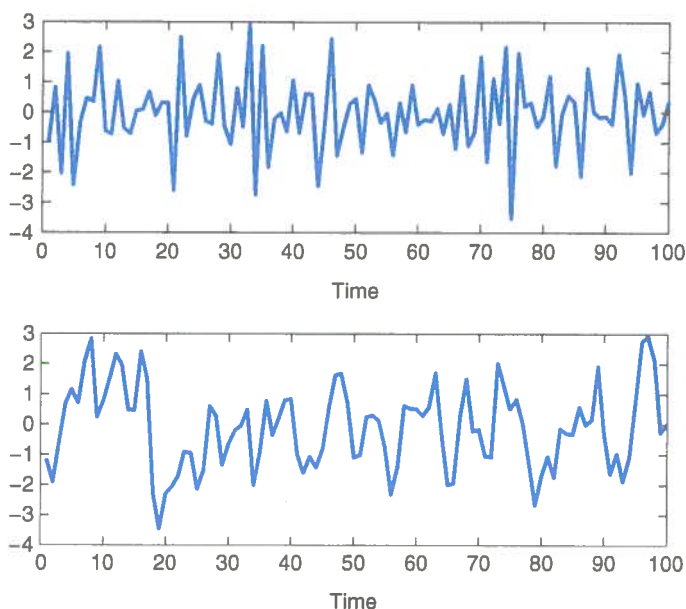


Figure 3.17 Realizations from MA(1) processes defined by (3.90) with $q = 1$. **Top panel:** $\beta = -0.7$. **Bottom panel:** $\beta = 0.7$.

which gives an infinite-order AR process. This converges in mean square if $|\beta| < 1$. However, if we do the same back-substitution for (3.94), the sum does not converge for $|\beta| < 1$. Clearly, an MA(1) cannot always be represented as an infinite-order AR process. We say that the MA(1) given by (3.93) is *invertible* if it can be written as an infinite-order autoregressive process, which occurs if $|\beta| < 1$.

Figure 3.17 shows two realizations from an MA(1) process with different values of β . In the top panel, $\beta = -0.7$, showing more high-frequency behavior. In the bottom panel, $\beta = 0.7$, showing more low-frequency behavior, corresponding to the spectral densities shown in Figure 3.16.

Parameter Estimation

Estimation for the parameters of MA processes is not as straightforward as for those of AR processes, since autocovariances of MA processes are nonlinear in the parameters. One can still use the Yule–Walker equations to obtain estimators, but such estimators are inefficient (e.g., see Brockwell and Davis, 1991, Section 8.5). Just as the Durbin–Levinson algorithm can be used to obtain recursive estimators of AR parameters, the *innovations algorithm* can be used to obtain recursive moment-based estimators of MA parameters (e.g., Brockwell and Davis, 1991, Section 8.3), but they are not in general consistent. For these reasons, maximum-likelihood procedures are typically preferred for estimating MA parameters. However, given the aforementioned nonlinearity in the parameters, explicit solutions are generally not available and the likelihood

must be maximized numerically, such as by the Gauss–Newton algorithm (e.g. Shumway and Stoffer, 2006, Section 3.6). Note that it is often reasonable to use the results from the innovations algorithm as starting values in the numerical maximum-likelihood estimation procedure.

3.4.5 Autoregressive Moving-Average Process

We note the duality between the AR and the MA processes described above. That is, an $AR(p)$ process can be written as an infinite-order MA process and an $MA(q)$ process can be written as an infinite-order AR process. This suggests that a more parsimonious model for many processes might result from a combination of AR and MA processes. The time series $\{Y_t\}$ is said to be an *autoregressive moving-average (ARMA) process of order (p, q)* , or $ARMA(p, q)$, if

$$Y_t = \alpha_1 Y_{t-1} + \cdots + \alpha_p Y_{t-p} + W_t + \beta_1 W_{t-1} + \cdots + \beta_q W_{t-q}, \quad (3.96)$$

where $\alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q$ are fixed but unknown parameters, and $\{W_t\}$ is a white-noise process with zero mean and variance σ_w^2 . In the spatio-temporal setting, the analogous processes are called STARMAs (Section 6.4.2).

The properties of $ARMA(p, q)$ processes and estimation of their parameters can be found in most time series textbooks (e.g., see Shumway and Stoffer 2006, Section 3.2). In addition, many other variations of these models, including, for example, seasonality and differencing, form the foundations of modern parametric time series modeling (Box and Jenkins, 1970; Box, Jenkins, and Reinsel, 2008).

3.4.6 Graphical Models for Time Series

In Section 2.4, we presented directed and undirected graphs as a way to visualize conditional independencies in a joint distribution. Consider this approach applied to time series models for $\{Y_r : r \in D_t\}$. Figure 3.18a shows the directed graph associated with a first-order autoregressive ($AR(1)$) process and Figure 3.18b shows the directed graph associated with a second-order autoregressive ($AR(2)$) process.

In both graphs, consider node t and condition on values associated with all nodes that have arrows between them and t , here node $(t-1)$ for Figure 3.18a and nodes $\{(t-2), (t-1)\}$ for Figure 3.18b. Upon removing these nodes, each respective graph breaks into two unconnected graphs. Using expression (2.20) we see that for the $AR(1)$ time series (Figure 3.18a) we have

$$[Y_t | Y_{t-1}, Y_{t-2}, Y_{t-3}, \dots] = [Y_t | Y_{t-1}],$$

and for the $AR(2)$ time series (Figure 3.18b) we have

$$[Y_t | Y_{t-1}, Y_{t-2}, Y_{t-3}, \dots] = [Y_t | Y_{t-1}, Y_{t-2}].$$

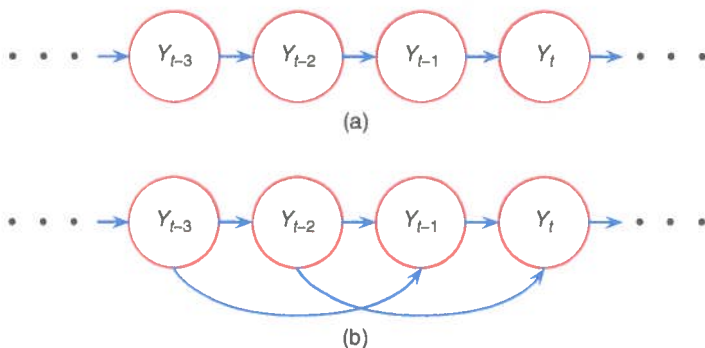


Figure 3.18 (a) Directed graph representing an AR(1) process. (b) Directed graph representing an AR(2) process.

From the directed graphs in Figure 3.18, it is clear that these time series possess a “lack-of-memory” or *Markov*-type property, in that the distribution of the current state, conditional on the past, in fact only depends on the “recent” past.

There are other ways to express temporal dependence that cannot be represented succinctly through a directed graph. For example, the first-order moving average (MA(1)) time series,

$$Y_t = \varepsilon_t + \varepsilon_{t-1},$$

where $\{\varepsilon_t\}$ are iid random variables, has a conditional distribution $[Y_t | Y_{t-1}, Y_{t-2}, Y_{t-3}, \dots]$ that depends on *all* the past values. However, notice that unconditionally there are some very simple *independencies* for the MA(1) time series; for example, $[Y_t, Y_{t+2}] = [Y_t][Y_{t+2}]$.

Another way to express temporal dependence is essentially to ignore the directionality of time and specify only the first two moments of $\{Y_t\}$, namely $E(Y_t)$ and $\text{cov}(Y_t, Y_r)$, for all t and r in the temporal domain of interest D_t . This approach is discussed in some detail in Section 3.3.

3.4.7 Vector Autoregressive Process

It is unusual to find time series in the real world that behave independently of other processes. Thus, it is important to consider several time series simultaneously, which are referred to as *multivariate time series*. As we shall see in Section 6.4, in the spatio-temporal setting it is sometimes useful to think of time series at different spatial locations as multiple (related) time series. We provide here a minimal introduction to some aspects of multivariate discrete time series, and we refer the reader to more advanced treatments (e.g., Lütkepohl, 1993, 2005) for additional discussion and detail.

As with univariate models, there are simple multivariate temporal processes that are useful for modeling many real-world multivariate time series. In this

regard, there are analogous models for multivariate white-noise, random walk autoregressive, moving-average, and ARMA processes. Given its importance to spatio-temporal modeling, we focus here on the *vector autoregressive (VAR) process*.

The VAR process relates the values of the multiple time series at a time t to some linear combination of the multiple time series at previous times. Thus analogous to the univariate AR case, for a K -variate time series, the VAR(p) process is given by

$$\mathbf{Y}_t = \mathbf{M}_1 \mathbf{Y}_{t-1} + \mathbf{M}_2 \mathbf{Y}_{t-2} + \cdots + \mathbf{M}_p \mathbf{Y}_{t-p} + \mathbf{W}_t, \quad (3.97)$$

where $\mathbf{Y}_t \equiv (Y_t^{(1)}, \dots, Y_t^{(K)})'$, $\{\mathbf{M}_l: l = 1, \dots, p\}$ are $K \times K$ propagator matrices (sometimes referred to as state-transition matrices), and $\{\mathbf{W}_t\}$ is a K -variate white-noise process with mean zero and covariance matrix $\mathbf{Q}$. Typically, this white-noise process is assumed to be Gaussian, in which case $\mathbf{W}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q})$.

Specifically, for illustration, consider the VAR(1) process,

$$\mathbf{Y}_t = \mathbf{M} \mathbf{Y}_{t-1} + \mathbf{W}_t. \quad (3.98)$$

Very simply, the i th row of the transition matrix $\mathbf{M}$ contains weights that describe how the multiple time series at time $t-1$ are linearly combined to influence the i -th time series at time t . For example,

$$Y_t^{(i)} = m_{i,1} Y_{t-1}^{(1)} + m_{i,2} Y_{t-1}^{(2)} + \cdots + m_{i,K} Y_{t-1}^{(K)} + W_t^{(i)}, \quad (3.99)$$

where $m_{i,j}$ refers to the (i, j) th entry in $\mathbf{M}$, and $W_t^{(i)}$ refers to the i th element of the vector $\mathbf{W}_t$; $i = 1, \dots, K$. Of course, it is precisely this interaction between the K time series, and between the K potentially correlated noise processes, that can lead to complicated behavior. One can show that the VAR(1) process is causal if the eigenvalues of $\mathbf{M}$ are all less than one in modulus.

Parameter Estimation for the VAR(1) Process

Estimation of the parameter matrices $\mathbf{M}$ and $\mathbf{Q}$ can be obtained via maximum-likelihood (see Lütkepohl, 1993, Chapter 3) or method-of-moments. For illustration, we consider the latter. Assume that the process $\{\mathbf{Y}_t\}$ has mean zero and is second-order stationary; that is, all the autocovariance and cross-covariance functions depend only on the lag difference between temporal indices. We then post-multiply both sides of (3.98) by $\mathbf{Y}_{t-1}'$ and take expectations to obtain

$$E(\mathbf{Y}_t \mathbf{Y}_{t-1}') = \mathbf{M} E(\mathbf{Y}_{t-1} \mathbf{Y}_{t-1}'), \quad (3.100)$$

or

$$\mathbf{C}_Y^{(1)} = \mathbf{M} \mathbf{C}_Y^{(0)}, \quad (3.101)$$

where we define $\mathbf{C}_Y^{(\tau)}$ to be the τ th lagged covariance matrix of the process $\{\mathbf{Y}_t\}$. Thus, we solve for $\mathbf{M}$ and obtain

$$\mathbf{M} = \mathbf{C}_Y^{(1)} (\mathbf{C}_Y^{(0)})^{-1}. \quad (3.102)$$

Another useful relationship that expresses $\mathbf{C}_Y^{(0)}$ directly in terms of $\mathbf{M}$ and $\mathbf{Q}$ is (Harvey, 1993, Section 7.3)

$$\text{vec}(\mathbf{C}_Y^{(0)}) = [\mathbf{I} - \mathbf{M} \otimes \mathbf{M}]^{-1} \text{vec}(\mathbf{Q}), \quad (3.103)$$

where the symbol $\otimes$ denotes the Kronecker product and the $\text{vec}(\cdot)$ operator stacks the columns of an $a \times b$ matrix to make an $ab \times 1$ column vector (e.g., see Lütkepohl, 1993, for more details).

Then, assuming we have observations $\mathbf{Y}_1, \dots, \mathbf{Y}_T$, the empirical lagged covariance matrices are

$$\hat{\mathbf{C}}_Y^{(\tau)} \equiv \frac{1}{T} \sum_{t=1}^{T-\tau} (\mathbf{Y}_{t+\tau} - \hat{\boldsymbol{\mu}})(\mathbf{Y}_t - \hat{\boldsymbol{\mu}})', \quad (3.104)$$

where the K -dimensional empirical mean is, $\hat{\boldsymbol{\mu}} \equiv \frac{1}{T} \sum_{t=1}^T \mathbf{Y}_t$. As with the univariate case, the divisor T in (3.104) avoids problems with positive-definiteness, but the divisor $(T - \tau)$ mitigates the obvious bias resulting from using T .

It follows from (3.102) that the method-of-moments estimator of $\mathbf{M}$ is given by

$$\hat{\mathbf{M}} = \hat{\mathbf{C}}_Y^{(1)} (\hat{\mathbf{C}}_Y^{(0)})^{-1}, \quad (3.105)$$

where $\hat{\mathbf{C}}_Y^{(0)}$ must be nonsingular. Clearly, it is singular if $T < K$. Although this is not typically the case in multivariate time series analysis, it does become a critical issue in dynamic spatio-temporal models (see Chapter 7). Finally, note that although a zero mean vector was assumed in arriving at (3.105) via (3.104), the estimator $\hat{\mathbf{M}}$ is equally appropriate for $\boldsymbol{\mu} \neq \mathbf{0}$, since there is a correction, $\hat{\boldsymbol{\mu}}$, for $\boldsymbol{\mu}$ in (3.104).

One can obtain an estimator of $\mathbf{Q}$, the white-noise (in time) covariance matrix, in a similar manner. Post-multiply (3.98) by $\mathbf{Y}_t'$ and take expectations of both sides to obtain $\mathbf{Q} = \mathbf{C}_Y^{(0)} - \mathbf{M}\mathbf{C}_Y^{(1)'}.$ A method-of-moments estimator is then given by

$$\hat{\mathbf{Q}} = \hat{\mathbf{C}}_Y^{(0)} - \hat{\mathbf{M}}\hat{\mathbf{C}}_Y^{(1)'} \quad (3.106)$$

$$= \hat{\mathbf{C}}_Y^{(0)} - \hat{\mathbf{C}}_Y^{(1)} (\hat{\mathbf{C}}_Y^{(0)})^{-1} \hat{\mathbf{C}}_Y^{(1)'}. \quad (3.107)$$

Maximum-likelihood estimation must generally proceed by numerical maximization of the loglikelihood (e.g., Lütkepohl, 1993). However, if one makes the assumption that $\mathbf{Y}_0$ is known, then maximization of the conditional loglikelihood (conditional on $\mathbf{Y}_0$) can be done analytically. In that case, the maximum-likelihood and the least-squares estimators are equivalent (e.g. Harvey 1993, Section 7.4).

3.4.8 Nonlinear Time-Series Models

As discussed in Section 3.2.2, the behavior of many real-world processes can be characterized by nonlinear dynamical systems. Although it is fairly easy to distinguish between linear and nonlinear processes in the case of deterministic processes, it is not always so easy to distinguish between linear and nonlinear *stochastic* processes. In the case of deterministic processes, nonlinearity is fairly easily characterized by nonlinear interactions in the lagged process state variable. However, in the stochastic setting, nonlinearity can occur from this type of behavior, as well as from parameter interactions and error-process interactions (with white-noise or colored-noise processes and/or state processes.) Furthermore, the characteristics associated with non-Gaussian errors and/or temporal nonstationarity are often indistinguishable from characteristics of nonlinearity (e.g., see Chatfield, 2004, Section 11.1.2 for an excellent overview.) Even with this lack of a clear definition of nonlinearity, there are several model classes in the time-series literature that are commonly viewed as nonlinear. We shall discuss briefly a few of these to facilitate the presentation of spatio-temporal nonlinearity in Chapter 7.

A very general class of nonlinear autoregressive models for the process $\{Y_t\}$ can be written as

$$Y_t = \mathcal{M}(Y_{t-1}, \dots, Y_{t-p}) + \tau(Y_{t-1}, \dots, Y_{t-p})\epsilon_t, \quad t = p, p+1, \dots, \quad (3.108)$$

where $\{\epsilon_t\}$ are *iid* with mean 0 and variance 1, and $\mathcal{M}$ and τ are functions of the lagged state process. In general, these functions can be specified parametrically or nonparametrically and can accommodate many different univariate nonlinear time series models as special cases [e.g., see Fan and Yao (2005, Section 1.5.4) for a complete discussion]. We consider briefly a few of these models here, for the special case where $p = 1$. Details and more general forms for these models can be found in the references listed in the Bibliographic Notes in Section 3.7.

Consider the (deceptively simple) first-order autoregression

$$Y_t = \mathcal{M}(Y_{t-1}) + \epsilon_t, \quad t = 1, 2, \dots, \quad (3.109)$$

where $\{\epsilon_t\}$ are *iid* with mean 0 and variance σ_ϵ^2 , $\mathcal{M}$ is some *nonlinear function* of Y_{t-1} , and one can specify various distributions for ϵ_t . Such a model is often

referred to as a *functional coefficient autoregressive (FAR)* model (e.g., Fan and Yao, 2005, Section 1.5.4). There are various approaches to specifying the model for $\mathcal{M}$, from fully nonparametric to fully parametric.

Another way to accommodate nonlinearity in autoregressive models is to allow parameters in the linear autoregressive model to vary (randomly) as a function of time. That is,

$$Y_t = a_t Y_{t-1} + \epsilon_t, \quad t = 1, 2, \dots, \quad (3.110)$$

where $\{\epsilon_t\}$ are the same as in (3.109). There are several different approaches to modeling this time-varying autoregressive coefficient, depending on what exogenous and endogenous information is available and whether one views a_t as random or not. For example, say there is another (exogenous) process $\{X_t : t = 1, 2, \dots\}$ available. One might consider the system

$$Y_t = a_t Y_{t-1} + \epsilon_t, \quad \epsilon_t \sim iid(0, \sigma_\epsilon^2), \quad (3.111)$$

$$a_t = b_0 + b_1 X_t + \gamma_t, \quad \gamma_t \sim iid(0, \sigma_\gamma^2), \quad (3.112)$$

where $\{\epsilon_t\}$ and $\{\gamma_t\}$ are independent, with distributions unspecified. In this case, the nonlinearity is apparent after substituting (3.112) into (3.111) [e.g., see Chatfield (2004, Section 11.2)]. A plot of Y_t versus Y_{t-1} would show both the autoregressive coefficient and the variance changing with time t . Clearly, more complicated relationships (nonlinear, semiparametric) between X_t and a_t could be specified, and other exogenous variables could be considered. In addition, one might consider $\{X_t\}$ to be a random process. Now, if covariates (or exogenous processes) were not available or considered unimportant, one could consider a simple stochastic time series model for a_t in the system:

$$Y_t = a_t Y_{t-1} + \epsilon_t; \quad \epsilon_t \sim iid(0, \sigma_\epsilon^2), \quad (3.113)$$

$$a_t = b_0 + b_1 a_{t-1} + \gamma_t; \quad \gamma_t \sim iid(0, \sigma_\gamma^2), \quad (3.114)$$

where $\{\epsilon_t\}$ and $\{\gamma_t\}$ are independent. Once again, more complicated models could be considered for $\{a_t\}$. Note that both pairs of equations, (3.111), (3.112) and (3.113), (3.114), are hierarchical representations. If $\{Y_t\}$ is not directly observed (i.e., latent), then a data model could be added to this hierarchy. In addition, the parameters in the models for $\{a_t\}$ could be considered random as well, and time-varying variances could also be included.

A related system assumes that the AR coefficient in (3.110) changes in a more discrete fashion, depending on the value of some exogenous variable, or the present (or past) values of the state itself (Tong, 1990). For example, a simple *threshold autoregression* model can be written as

$$Y_t = \begin{cases} a^{(1)} Y_{t-1} + \epsilon_t^{(1)}; & \epsilon_t^{(1)} \sim iid(0, \sigma_1^2), & \text{for } X_t \leq c, \\ a^{(2)} Y_{t-1} + \epsilon_t^{(2)}; & \epsilon_t^{(2)} \sim iid(0, \sigma_2^2), & \text{for } X_t > c, \end{cases} \quad (3.115)$$

where the model can be extended in an obvious way to account for more than one condition (threshold), c . Thus, there are different linear AR models depending on how the exogenous variable X_t compares to the corresponding threshold(s). The *state-dependent* version of this threshold autoregression model substitutes Y_{t-1} (or, in general, a function of past values $Y_{t-1}, \dots, Y_{t-p}$ for X_t , so that the current model depends on some past value of the state process. Now, it could be that X_t is not actually observed, in which case the different regimes are based on a latent process. In addition, one may not actually know the thresholds associated with the regime transition, and thus these might need to be estimated or allowed to be random as well. In general, these models (referred to as *regime-switching* models) can be quite complicated!

As mentioned above, it is certainly possible that the variances in time series models can vary with time as well. In general, models for time-varying variances can be parametric, semiparametric, or nonparametric and can be deterministic or stochastic. In the deterministic parametric context, the classical formulation is the *generalized autoregressive conditional heteroskedasticity* (GARCH) model,

$$\sigma_t^2 = a_0 + \sum_{i=1}^p a_i Y_{t-i}^2 + \sum_{j=1}^q b_j \sigma_{t-j}^2, \quad (3.116)$$

where $a_i, b_j \geq 0$, for all i, j . These models are commonly used in the time series and econometrics literatures. One may also consider a stochastic variance process (i.e., *stochastic volatility*) in which case σ_t^2 (or σ_t) is modeled as a stochastic process. For example, one such (simple) model is given by

$$\log \sigma_t^2 = b_0 + b_1 \log \sigma_{t-1}^2 + \eta_t, \quad \eta_t \sim iid(0, \sigma_\eta^2), \quad (3.117)$$

where $\{\eta_t\}$ is independent of the error process $\{\varepsilon_t\}$ in (3.109). Of course, (3.117) may be generalized to have higher-order lags, and the autoregressive function can be nonparametric. We note that the stochastic-volatility formulation fits naturally into the hierarchical statistical modeling framework.

In principle, all of the nonlinear models discussed here can be extended to the multivariate setting (e.g., see Fan and Yao, 2005). Such models rapidly begin to suffer from the “curse of dimensionality,” as the number of variables in the state process increases. This makes estimation difficult in general and typically, one must specify very specific forms for these models. We shall return to this issue when we discuss nonlinear spatio-temporal models in Chapter 7.

3.5 SPECTRAL REPRESENTATION OF TEMPORAL PROCESSES

It is often useful to represent a time series in terms of projections onto a set of underlying spectral basis functions. There are several reasons why we might want to do this. We might be interested in partitioning the overall variability

in a time series into different scales of variability. For example, how much variability in the daily-temperature time series at a location can be described by an annual cycle or a semi-annual cycle? In addition, what quasi-periodic phenomena account for the most variability in a physical system? Another reason for thinking about time series projections onto basis functions is that such projections often have much simpler correlation structure than the original series. That is, the transformation into the spectral domain often acts as a decorrelator. It is also the case that some of the projections do not account for much variability and thus can often be ignored without much loss of information. This suggests that lower-dimensional representations of the process can be obtained in the spectral domain than can be obtained in the time domain.

In what follows, we consider a general spectral representation of temporal processes. We do not present many of the technical details; rather, we refer the reader to one of the many more technical books on times series and their spectral representation listed in the Bibliographic Notes in Section 3.7.

3.5.1 Spectral Representations of Continuous-Time Functions via Orthogonal Series

Consider a process that can be written as a function of time, $f(t)$, over the time interval (a, b) . We define a sequence of spectral basis functions $\phi_k(t)$; $k = 1, 2, \dots$, to be *orthogonal* over the interval (a, b) if

$$\int_a^b \phi_k(t) \phi_l(t) dt = \begin{cases} 0, & k \neq l, \\ \delta, & k = l, \end{cases} \quad (3.118)$$

where $\delta > 0$. Furthermore, these functions are said to be *orthonormal* if $\delta = 1$.

We now seek to expand $f(t)$ in terms of these functions:

$$f(t) = \sum_{k=1}^{\infty} \alpha_k \phi_k(t), \quad (3.119)$$

where $\{\alpha_k\}$ are the “weights” or *spectral coefficients* of the spectral expansion. If this expansion is possible, then $\{\alpha_k\}$ are given by

$$\alpha_k = \int_a^b f(t) \phi_k(t) dt, \quad k = 1, 2, \dots \quad (3.120)$$

That is, the spectral coefficients are simply the projection of $f(t)$ onto the orthogonal basis functions.

In some cases, we may be interested in truncating (3.119) to consider only up to the p_α th function. The *mean squared error* (MSE) of the truncation is then given by

$$\text{MSE} \equiv \int_a^b \left(f(t) - \sum_{k=1}^{p_\alpha} \alpha_k \phi_k(t) \right)^2 dt. \quad (3.121)$$

It can be shown that the MSE is minimized with respect to $\{\alpha_k : k = 1, 2, \dots, p_\alpha\}$, by the coefficients (3.120).

In addition to the orthogonality property, it is useful to consider one additional property of the basis functions, namely *completeness* (sometimes called *L_2 -completeness*). The orthogonal series $\{\phi_k(t)\}$ is complete over the interval (a, b) if, for every piecewise continuous function $f(t)$ on that interval, the MSE (3.121) tends to zero as $p_\alpha \rightarrow \infty$ [e.g., see Kaplan (1991, Section 7.11)]

There are many examples of orthogonal basis functions that can be used for temporal processes. Common bases include trigonometric (i.e., Fourier) wavelet, spline, and orthogonal polynomial functions. In the next section, we illustrate the trigonometric case in some detail.

Trigonometric Series Expansion

Consider a function of interest, $f(t)$, defined on the interval $(-1/2, 1/2)$. Let $\{\phi_k(t)\}$ correspond to the trigonometric basis functions, $\sin(2\pi kt)$ and $\cos(2\pi kt)$, for $k = 0, 1, 2, \dots$. These basis functions are complete and orthogonal. Hence, we can write the function $f(t)$ as

$$f(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \{a_k \cos(2\pi kt) + b_k \sin(2\pi kt)\}, \quad (3.122)$$

and the *Fourier coefficients* are

$$a_k = 2 \int_{-1/2}^{1/2} f(t) \cos(2\pi kt) dt, \quad k = 0, 1, \dots, \quad (3.123)$$

$$b_k = 2 \int_{-1/2}^{1/2} f(t) \sin(2\pi kt) dt, \quad k = 1, \dots. \quad (3.124)$$

Note that we can make use of Euler's relationship, $e^{\pm i 2\pi kt} \equiv \cos(2\pi kt) \pm i \sin(2\pi kt)$, where $i = \sqrt{-1}$, to simplify the notation (see Section 3.2.1). In this case, we can write (3.122) in its common form:

$$f(t) = \sum_{k=-\infty}^{\infty} \alpha_k e^{i 2\pi kt}, \quad -1/2 \leq t \leq 1/2, \quad (3.125)$$

where $\phi_k(t) \equiv e^{i 2\pi kt}$ and $\alpha_k \equiv a_k + i b_k$ are both complex functions. The result (3.125) is, of course, just the Fourier transform given in (3.66), and it is known as the *spectral representation theorem* (see Section 3.3).

The coefficients a_k and b_k provide important information regarding the characteristics of the sinusoidal time series, as shown in Section 3.2.1. It follows that when considering (3.122), the coefficients a_k and b_k , $k = 0, 1, 2, \dots$, have associated amplitude and phase given by $\sqrt{a_k^2 + b_k^2}$ and $\tan^{-1}(-b_k/a_k)$, respectively.

3.5.2 Discrete-Time Spectral Expansion

The trigonometric spectral representation can be applied in the discrete-time setting (i.e., for time series). Let $\{Y_t : t = 1, \dots, T\}$ be a time series and define $\{\phi_k(t) : t = 1, \dots, T ; k = 1, \dots, p_\alpha\}$ to be a complete set of basis functions. Define $\mathbf{Y} \equiv (Y_1, \dots, Y_T)'$, $\boldsymbol{\alpha} \equiv (\alpha_1, \dots, \alpha_{p_\alpha})'$, and $\boldsymbol{\Phi} \equiv (\phi_1, \dots, \phi_{p_\alpha})$, where $\phi_k \equiv (\phi_k(1), \dots, \phi_k(T))'$. Then, it is convenient to denote the spectral expansion,

$$Y_t = \sum_{k=1}^{p_\alpha} \alpha_k \phi_k(t), \quad (3.126)$$

as

$$\mathbf{Y} = \boldsymbol{\Phi} \boldsymbol{\alpha}. \quad (3.127)$$

Multiply both sides of (3.127) by $\boldsymbol{\Phi}$ to obtain the spectral-coefficient vector,

$$\boldsymbol{\alpha} = (\boldsymbol{\Phi}' \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}' \mathbf{Y}, \quad (3.128)$$

which is in the form of an ordinary-least-squares estimate. In the case where the basis functions are orthonormal, $\boldsymbol{\Phi}' \boldsymbol{\Phi} = \mathbf{I}$; then (3.128) becomes

$$\boldsymbol{\alpha} = \boldsymbol{\Phi}' \mathbf{Y}. \quad (3.129)$$

Such a projection of data onto discrete basis functions is historically known as *harmonic analysis*, especially in the case where the basis functions are Fourier basis functions. As we shall see, estimation based on (3.128) does not yield consistent estimates of the spectrum, but it does have utility with regard to dimension reduction and decorrelation.

As an example, consider the Fourier series expansion in the discrete-time setting with the series length T an even positive integer (this is not necessary, but makes the notation simpler). Note that the highest frequency we can fit to the data is that corresponding to two measurements per cycle. This is called the *Nyquist frequency*. For example, if we have measurements of temperature at noon each day, we know nothing about what is happening within any day. Thus, the highest frequency we can consider corresponds to one cycle every two days (i.e., 1/2 cycle per day). This is related to another issue in spectral analysis known as *aliasing*, which refers to the presence of incorrect frequencies simply

due to the sampling of a series at too slow a rate. For example, if the true-process frequency was 1/2 cycle per day, but we sampled say every third day, we would see (false) evidence of periodic behavior with a period of 6 days per cycle (i.e., a frequency of 1/6 cycle per day). Thus, one should sample a series at a rate of at least two measurements per cycle for the cycle corresponding to the highest conceivable frequency of interest.

In general, for discrete time series, we only need to consider the frequencies,

$$\omega_k = \frac{2\pi k}{T}, \quad k = 1, \dots, T/2, \quad (3.130)$$

which are in units of radians per sampling increment. Often, these frequencies are denoted in terms of *cycles* per sampling increment, in which case $\omega_k = k/T$, $k = 1, \dots, T/2$. In the former case, the Nyquist frequency is $\omega_{T/2} = \pi$ radians per sampling increment, and in the latter case it is 1/2 cycle per sampling increment. Then the orthonormal basis functions associated with α_k , for $k = 1, \dots, T/2 - 1$, are given by

$$\phi_k(t) = \sqrt{2/T} \left(\cos\left(\frac{2\pi k}{T}t\right) + i \sin\left(\frac{2\pi k}{T}t\right) \right), \quad t = 1, \dots, T. \quad (3.131)$$

In addition, $\phi_k(t)$ for $k = 0$ and $k = T/2$ are given by

$$\phi_0(t) = \sqrt{1/T}, \quad t = 1, \dots, T, \quad (3.132)$$

and

$$\phi_{T/2}(t) = \sqrt{1/T} \cos(\pi t), \quad t = 1, \dots, T. \quad (3.133)$$

Thus, for each of the $T/2 - 1$ basis functions given by (3.131), $\alpha_k = a_k + i b_k$ is complex and represents the contribution associated with the k th basis function (which contains k cycles over the series of length T). Some practitioners prefer to consider the analysis without the use of the complex-number notation. In that case, one can simply define T basis functions corresponding to a_k and b_k as follows:

$$\phi_{a_0}(t) \equiv \sqrt{1/T} \quad (3.134)$$

$$\phi_{a_k}(t) \equiv \sqrt{2/T} \cos\left(\frac{2\pi k}{T}t\right) \quad (3.135)$$

$$\phi_{b_k}(t) \equiv \sqrt{2/T} \sin\left(\frac{2\pi k}{T}t\right) \quad (3.136)$$

$$\phi_{a_{T/2}}(t) \equiv \sqrt{1/T} \cos(\pi t), \quad (3.137)$$

where $t = 1, \dots, T$ and $k = 1, \dots, T/2 - 1$; it is clear from the complex representation above that we do not need a basis function corresponding to b_0 or $b_{T/2}$; we can think of $b_0 = b_{T/2} = 0$. Thus, in this formulation, there are T basis functions, and they can be used to construct the $T \times T$ matrix, Φ , defined at the beginning of this subsection.

Consider the simple example where we have a time series of length $T = 6$ given by

$$\mathbf{Y} = (-.0831, .7187, .2182, .3068, -.9568, -.4170)'. \quad (3.138)$$

This series is plotted in Figure 3.19. It is evident from this plot that most of the variability is occurring at a frequency of approximately one cycle per six time increments. We can construct the following basis functions:

$$\phi_{a_0}(t) = \frac{1}{\sqrt{6}} \quad (3.139)$$

$$\phi_{a_1}(t) = \sqrt{2/6} \cos\left(\frac{2\pi}{6}t\right) \quad (3.140)$$

$$\phi_{b_1}(t) = \sqrt{2/6} \sin\left(\frac{2\pi}{6}t\right) \quad (3.141)$$

$$\phi_{a_2}(t) = \sqrt{2/6} \cos\left(\frac{4\pi}{6}t\right) \quad (3.142)$$

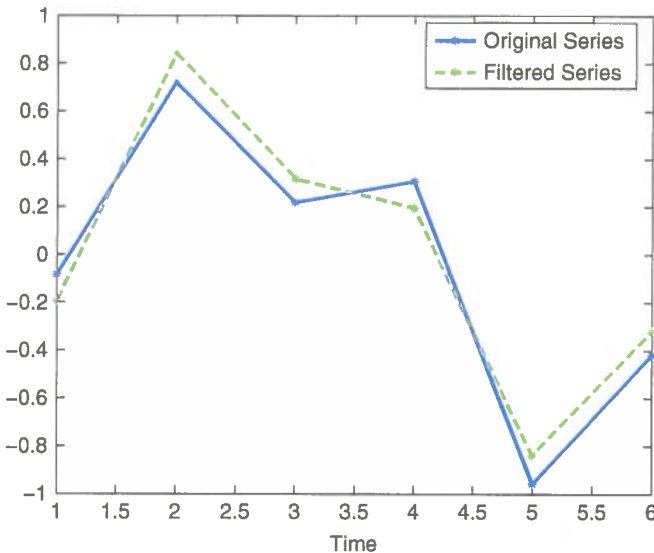


Figure 3.19 Original time series given by (3.138) (solid blue line) and filtered representation given by (3.139) (dashed green line).

$$\phi_{b_2}(t) = \sqrt{2/6} \sin\left(\frac{4\pi}{6}t\right) \quad (3.143)$$

$$\phi_{a_3}(t) = \sqrt{1/6} \cos(\pi t), \quad (3.144)$$

for $t \in \{1, 2, \dots, 6\}$. Letting $\Phi = [\phi_{a_0} \phi_{a_1} \phi_{b_1} \phi_{a_2} \phi_{b_2} \phi_{a_3}]$, we obtain

$$\Phi = \begin{bmatrix} .4082 & .2887 & .5000 & -.2887 & .5000 & -.4082 \\ .4082 & -.2887 & .5000 & -.2887 & -.5000 & .4082 \\ .4082 & -.5774 & .0000 & .5774 & .0000 & -.4082 \\ .4082 & -.2887 & -.5000 & -.2887 & .5000 & .4082 \\ .4082 & .2887 & -.5000 & -.2887 & -.5000 & -.4082 \\ .4082 & .5774 & .0000 & .5774 & .0000 & .4082 \end{bmatrix}. \quad (3.145)$$

Write $\alpha = (\tilde{a}_0, \tilde{a}_1, \tilde{b}_1, \tilde{a}_2, \tilde{b}_2, \tilde{a}_3)'$, where

$$\alpha = \Phi'Y = (-.0871, -.9630, .6428, -.1106, .2309, .5839)'. \quad (3.146)$$

The amplitudes associated with the orthonormal eigenfunctions (3.139)-(3.144) are then given by $A_k \equiv \sqrt{\tilde{a}_k^2 + \tilde{b}_k^2}$, where $\tilde{b}_0 \equiv 0$. Hence,

$$A_0 = .09, \quad A_1 = 1.16, \quad A_2 = .26, \quad A_3 = .58. \quad (3.147)$$

Thus, the most dominant cycles (i.e., frequencies corresponding to the most variation) in this time series correspond to A_1 and A_3 , which represent one cycle per six time periods and three cycles per six time periods, respectively. We can always reconstruct the original time series from the spectral coefficients by the formula $Y = \Phi\alpha$. We can also reconstruct the time series after setting some of the less important coefficients in α equal to zero (i.e., *filtering*). For example, assume a threshold value corresponding to an amplitude of 0.5, so that any coefficients corresponding to an amplitude less than 0.5 are set to zero. Thus, we obtain the filtered time series, Y_R , from the vector,

$$\alpha_R = (0, -.9630, .6428, 0, 0, .5839)', \quad (3.148)$$

as follows:

$$Y_R \equiv \Phi\alpha_R = (-.1950, .8378, .3176, .1950, -.8378, -.3176)'. \quad (3.149)$$

Figure 3.19 shows the plot of this filtered time series along with the original time series. Most of the variability in the original series is still evident in the series with only the one-cycle and three-cycle components included. This illustrates that spectral approaches can be used for dimension reduction; in this case, we can characterize most of the variability with only half the number of parameters (i.e., $\tilde{a}_1$, $\tilde{b}_1$, and $\tilde{a}_3$). However, note that such a filtering approach

is quite naive, and more sophisticated approaches are used in practice (e.g., Shumway and Stoffer, 2006, Chapter 4).

In practice, one does not typically form the matrix Φ as shown in this example, since for time series of even moderate length, the dimensionality of Φ is prohibitive, from a computational and storage perspective. Fortunately, this matrix does not have to be formed directly. Rather, the *Fast Fourier Transform* (FFT) algorithm (Cooley and Tukey, 1965) can be used to obtain the spectral coefficients in a very efficient manner. Thus, the operation $\Phi'Y$ can be carried out using the FFT algorithm, and the operation $\Phi\alpha$ can be carried out by the inverse FFT algorithm. Similar computational efficiencies can be obtained for other basis functions. For example, the multiresolution algorithm of Mallat (1989) can be used when the basis functions are wavelets. However, there are examples where one still must form the matrix Φ and compute the matrix multiplications directly. Examples include empirical basis functions (e.g., see the discussion of EOFs in Section 5.3) or theoretical basis functions suggested by the normal modes of some natural system (e.g., Wikle et al., 2001).

In addition to dimension reduction, the spectral decomposition also has a decorrelating effect on the original time series. For example, if the process is second-order stationary, then in the spectral domain,

$$\text{var}(\alpha) \simeq \text{diag}(\mathbf{d}), \quad (3.150)$$

where $\mathbf{d}$ is a vector containing the variances of the spectral coefficients $\{\alpha_i\}$. That is, α_i and α_j are asymptotically independent (e.g., Shumway and Stoffer, 2006, Appendix C). This can simplify models dramatically. For example, assume that $Y \sim \text{Gau}(\mathbf{0}, \Sigma)$, $\{\phi_k(\cdot)\}$ are orthonormal, and we have $Y = \Phi\alpha$, or equivalently, $\alpha = \Phi'Y$. Thus, using basic properties of the multivariate Gaussian distribution, we obtain $\alpha \sim \text{Gau}(\mathbf{0}, \Phi'\Sigma\Phi)$. When Φ are Fourier basis functions and Y comes from a stationary times series, it follows that $\Phi'\Sigma\Phi \simeq \text{diag}(\mathbf{d})$. Hence, from a modeling perspective, it is much easier to work with the process α than the process Y , since its dependence structure is much simpler.

3.5.3 Univariate Spectral Analysis

In the case of discrete Fourier basis functions, the coefficients $\{\alpha_k\}$ in the expansion (3.126) can be written in terms of the frequencies $\{\omega_k\}$: $\alpha(\omega_k) \equiv \alpha_k$, where recall from (3.130) that $\omega_k = 2\pi k/T$; $k = 0, \dots, T/2$. In this case, the Fourier basis functions are complex, and (3.129) is modified to

$$\alpha(\omega_k) = \sum_{t=1}^T Y_t \bar{\phi}_k(t), \quad (3.151)$$

where recall that $\{Y_t : t = 1, \dots, T\}$ is a time series, and the overbar indicates the complex conjugate of the Fourier basis functions given in (3.131), (3.132),

and (3.133). We can then define the *periodogram* as the squared modulus of $\alpha(\omega_k)$:

$$\hat{I}(\omega_k) \equiv \alpha(\omega_k)\bar{\alpha}(\omega_k) = |\alpha(\omega_k)|^2, \quad k = 0, \dots, T/2, \quad (3.152)$$

which correspond to the “amplitudes” for the frequencies $\{\omega_k\}$ discussed in the example of the previous section. One can also define the periodogram as the discrete version of (3.66), with $C(\tau)$ replaced with the empirical autocovariance function (3.71). Thus, the periodogram can be thought of as an estimator of the spectrum of $\{Y_t\}$.

As an estimator of the spectrum, the periodogram has some problems. Although the periodogram is asymptotically unbiased for stationary processes for finite sample sizes there may in fact be large biases, and the periodogram is not a consistent estimator of the spectrum (e.g., Chatfield, 1989, Section 7.3.2). Thus, we seek an estimator that reduces bias as well as variance. Of course (as is always the case in statistical estimation), there is a trade-off between bias reduction and variance reduction. We discuss briefly below some approaches that seek to deal with this trade-off. Although the topic is too extensive to cover in detail here, the Bibliographic Notes in Section 3.7 include resources for such details.

When the bias is asymptotically zero, one obvious way to reduce it is to use more data. Typically, that is not a viable option. Another option for bias reduction is *data tapering* (or *data windowing*), where each Y_t is replaced with a weighted version, $Y_t w_t$, where the weights $\{w_t\}$ typically taper to zero at the endpoints of the data and are equal to one in the middle. For example, a common data window is the cosine bell taper (e.g., Bloomfield, 2000, Section 6.2),

$$w_t = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi(t - .5)}{T} \right) \right], \quad t = 1, \dots, T. \quad (3.153)$$

Such data tapering is most useful when there are large differences between the beginning and end portions of the time series, or when certain frequency components contain much more variance than adjacent frequencies (e.g., Shumway and Stoffer, 2006, Section 4.5). Alternatively, we may *pre-whiten* the data. In this case, we filter the data initially (e.g., consider $U_t \equiv Y_t - Y_{t-1}$), perform the spectral analysis on the filtered data $\{U_t\}$, and then *post-color* the spectral estimate by accounting for the spectrum of the filter. (Note that in the frequency domain, this is a simple multiplication for each frequency.) This is described, for example, in Percival and Walden (1993, Section 6.5).

There are several alternatives when it comes to variance reduction. A common approach is to smooth the periodogram, whereby a moving average of the periodogram is calculated (Daniell, 1946):

$$\hat{f}(\omega_k) \equiv \frac{1}{2P+1} \sum_{j=k-P}^{k+P} \hat{I}(\omega_j). \quad (3.154)$$

Clearly, the spectral estimate at frequencies within P frequencies from the ends of the spectrum cannot be obtained with this estimator. As this approach amounts to low-pass filtering of the periodogram, we could easily pick other averaging approaches (filters) as well, and typically this is considered in the context of windows applied to the autocovariance function [e.g., see Chatfield (1989, Section 7.6) for a concise discussion]. By smoothing the periodogram estimates (which are essentially uncorrelated), we reduce the variance by order $1/P$, but at the expense of introducing a bias (and correlation). Clearly, the choice of P is critical in this regard. Chatfield (1989, Section 7.4.4) suggests choosing a P on the order of $T/40$, but then some additional trial-and-error is needed to balance the desire for smoothness versus identification of sharp spectral peaks.

Another approach to averaging the periodogram was proposed by Bartlett (1948), and it consists of splitting the data into segments, calculating the periodogram for each segment, and then averaging these multiple periodograms at each frequency. Welch (1967) extended this approach so that the segments overlap, thereby reducing the bias. So-called *multitaper* methods, which use a family of tapers (typically, “discrete prolate spheroidal sequences”) have been shown to be a convenient way to manage effectively the trade-off between bias and variance reduction in nonparametric spectral estimation (Thomson, 1982; Percival and Walden, 1993). The statistical properties of the various estimators have been well studied; see the Bibliographic Notes in Section 3.7.

Finally, we note that a parametric approach can be taken to spectral estimation, whereby a model (typically an ARMA model) is specified for which we know the theoretical spectral density in terms of the model parameters. The model is fitted to $\{Y_1, \dots, Y_T\}$ (e.g., Sections 3.4.3, 3.4.4, and 3.4.5), and the parameter estimates are substituted into the theoretical spectral density to obtain the spectral-density estimate. For more discussion of this approach, see, for example, Percival and Walden (1993, Chapter 9).

Example: The foundation of the theory of equatorial waves in the atmosphere was elegantly described in the landmark paper of Matsuno (1966). Matsuno showed, using a simplified set of hydrodynamical equations, that propagating waves were theoretically possible in tropical regions. Shortly thereafter, such waves were discovered in upper-level (rawinsonde) data, and their study has been of significant interest in the atmospheric sciences ever since. As an example, consider data for the north/south component of the upper (70-hPa)-level wind measured by rawinsondes daily at Chuuk (formerly Truk) Island in the tropical Pacific from 1964 to 1994, as described in Wikle, Madden, and Chen (1997). A smoothed-periodogram spectral analysis was performed on these 11,323 observations, after application of a split-cosine bell taper (10% on each end; e.g., see Bloomfield, 2000, Section 6.2). The spectral estimate was obtained from (3.154) with smoothing parameter $P = 50$. For comparison, an AR(24) spectral estimate was also obtained and both estimates are shown in Figure 3.20. Note that there are several spectral peaks associated

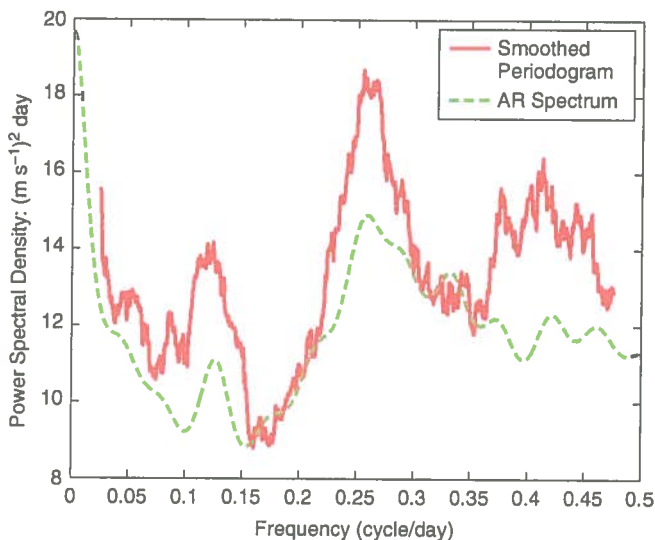


Figure 3.20 Spectral density (in units of $(m s^{-1})^2 day$) for the north/south component of the 70-hPa-level wind at Chuuk (formerly Truk) Island based on daily observations from 1964 to 1994. The smoothed periodogram estimate is given by the red solid line and the AR(24) spectral estimate is given by the dashed green line.

with these spectral estimates. Of particular interest, meteorologically, is the peak from about 0.22–0.33 cycles per day (corresponding to waves with periods of 3–4.5 days). As shown in Wikle, Madden, and Chen (1997), these are likely to correspond to the so-called “mixed-Rossby gravity waves” suggested by Matsuno’s theory and originally discovered in observations by Yanai and Maruyama (1966).

3.5.4 Bivariate Spectral Analysis

Recall that for jointly second-order stationary processes, $\{Y_t^{(1)}\}$ and $\{Y_t^{(2)}\}$, the cross-covariance function is given by $C_{12}(\tau)$, as defined in (3.68). If $\sum_{\tau=-\infty}^{\infty} |C_{12}(\tau)| < \infty$, then the *cross-spectrum* is defined as

$$f_{12}(\omega) \equiv \sum_{\tau=-\infty}^{\infty} C_{12}(\tau) \exp\{-2\pi i \omega \tau\}, \quad (3.155)$$

$$\equiv c_{12}(\omega) - i q_{12}(\omega), \quad -1/2 \leq \omega \leq 1/2, \quad (3.156)$$

where $f_{12}(\omega)$ is a complex-valued function, the real part, $c_{12}(\omega)$, is known as the *co-spectrum*, and the imaginary part, $q_{12}(\omega)$, is known as the *quadrature spectrum*. We can then define two useful quantities. The first is the *squared coherence function* (sometimes known as the *magnitude squared coherence*), which is given by

$$\text{coh}_{12}^2(\omega) \equiv \frac{|f_{12}(\omega)|^2}{f_1(\omega)f_2(\omega)} = \frac{c_{12}^2(\omega) + q_{12}^2(\omega)}{f_1(\omega)f_2(\omega)}, \quad -1/2 \leq \omega \leq 1/2, \quad (3.157)$$

where $f_i(\omega)$ refers to the spectral density of $\{Y_t^{(i)}\}$; $i = 1, 2$. The squared coherence is analogous to the usual squared correlation coefficient, in that $0 \leq \text{coh}_{12}^2(\omega) \leq 1$; it measures the square of the linear association between the components of the two processes at frequency ω .

In addition, we can define the *phase spectrum*, which is given by

$$\theta_{12}(\omega) \equiv \tan^{-1} \left(\frac{-q_{12}(\omega)}{c_{12}(\omega)} \right), \quad -1/2 \leq \omega \leq 1/2, \quad (3.158)$$

and varies between $-\pi$ and π . In conjunction with the squared coherence function, the phase spectrum is useful for establishing linear relationships between components of time series in the frequency domain. In particular, a constant phase together with a peak in the squared coherence suggests that one series leads/lags the other by the amount given by the phase. The example given at the end of this subsection will demonstrate how this works for bivariate time series data.

Let $\{\alpha^{(i)}(\omega_k)\}$ be the discrete Fourier transform (3.151) of series $\{Y_t^{(i)}\}$. Then, the *cross-periodogram* is defined as

$$I_{12}(\omega_k) \equiv \alpha^{(1)}(\omega_k) \bar{\alpha}^{(2)}(\omega_k), \quad (3.159)$$

where again the overbar indicates the complex conjugate. As an estimate of the cross-spectrum, the cross-periodogram suffers from the same problems (i.e., bias and lack of consistency) as the univariate periodogram. As in (3.154), we could smooth across frequencies:

$$\hat{f}_{12}(\omega_k) \equiv \frac{1}{2P+1} \sum_{j=k-P}^{k+P} \hat{I}_{12}(\omega_j). \quad (3.160)$$

Given this estimate of the cross-spectrum and estimates of the individual spectra, $\hat{f}_1(\omega_k)$ and $\hat{f}_2(\omega_k)$, one can obtain estimates of the co-spectrum and quadrature spectrum, the magnitude squared coherence, and the phase spectrum by simply substituting these into their respective definitions, (3.156), (3.157), and (3.158). The statistical properties of these estimators are well known and are described in the Bibliographic Notes in Section 3.7.

Example: Continuing the example presented in Section 3.5.3, we consider the east/west upper level (70-hPa) wind component at Chuuk Island, in addition to the north/south component. Figure 3.21 shows the smoothed

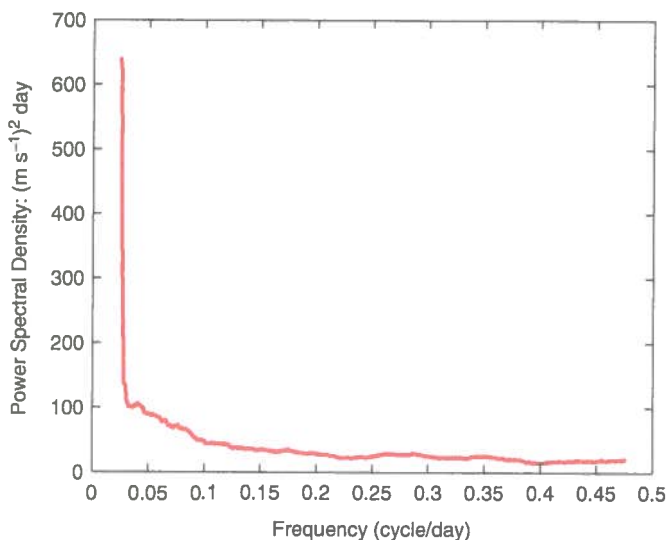


Figure 3.21 Spectral density (in units of $(m s^{-1})^2 day$) for the east/west component of the 70 hPa-level wind at Chuuk (formerly Truk) Island based on daily observations from 1964 to 1994. The smoothed periodogram estimate is shown, analogous to that for the north/south component shown in Figure 3.20.

periodogram estimate of the spectrum of this east/west wind component. Note that this spectrum is much more “red” than that for the north/south component shown in Figure 3.20; that is, it has much more power at lower frequencies. Clearly, the mixed-Rossby gravity wave frequencies are not as dominant in the east/west-wind-component observations. However, this does not mean they are not present. Consider the cross-spectral analysis of these two wind components. In particular, Figure 3.22 shows the magnitude squared coherence function and the phase spectrum estimates (using the same smoothed periodogram estimate as for Figure 3.21). There are substantial peaks in the squared coherence across several frequency bands. In particular, note that in the mixed-Rossby gravity wave frequency band (0.22–0.33 cycles per day) mentioned in Section 3.5.3 there is strong coherence, and the phase is close to $-\pi/2$ in this range, indicating that the north/south wind component leads the east/west wind component by a quarter of a cycle. This is precisely what was predicted by Matsuno’s (1966) theory for such waves. Wikle, Madden, and Chen (1997) discuss the likely wave types associated with other parts of the spectrum using a variety of cross-spectral analyses and multiple tropical stations (see also Section 5.2).

3.6 HIERARCHICAL MODELING OF TIME SERIES

As discussed in Chapter 2, hierarchical modeling (HM) is just the re-expression of a joint distribution of data, process, and parameters into a series of conditional models. Recall from basic probability that

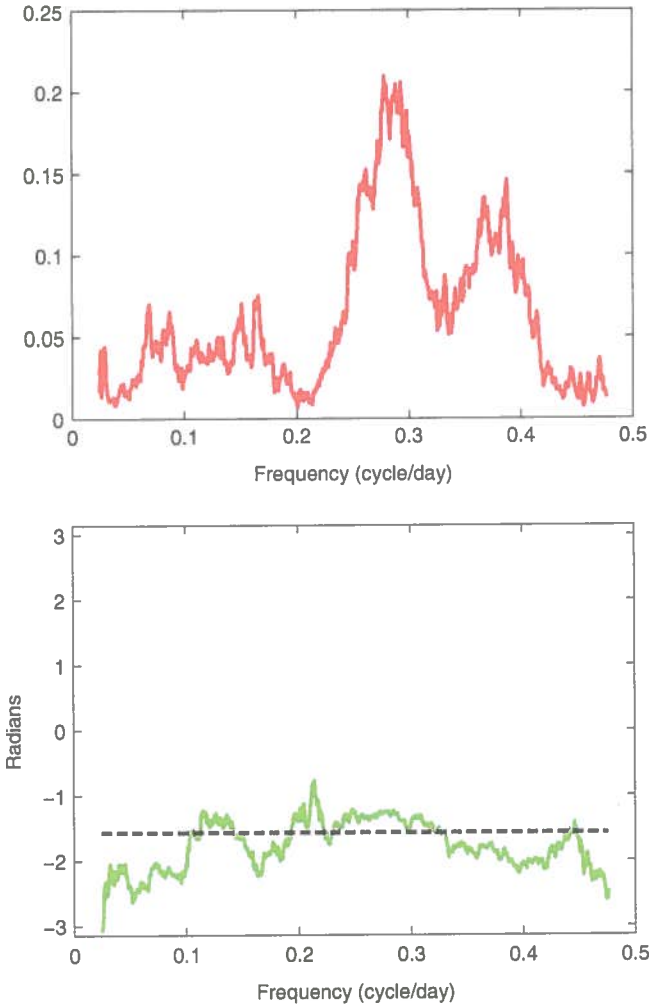


Figure 3.22 Cross-spectral analysis of the 70-hPa-level wind components at Chuuk (formerly Truk) Island based on daily observations from 1964 to 1994. **Top panel:** Estimate of the magnitude squared coherence function between the east/west and north/south wind components obtained from the smoothed periodogram estimates. **Bottom panel:** Estimate of the phase spectrum (in radians) associated with the magnitude squared coherence estimates in the top panel. The dashed black line is a reference line associated with a frequency of $-\pi/2$ radians. Note that a negative phase indicates that the north/south wind component leads the east/west component.

$[A, B, C] = [A|B, C][B|C][C]$. That is, the joint distribution of (say) three random variables, A, B, C , can always be factored into two conditional distributions and a marginal distribution. Thus, for complicated models for which there are data, a (hidden) process, and parameters involved, all known with more-or-less uncertainty, we consider the following general, three-stage factorization of [data, process, parameters] suggested by Berliner (1996):

Data model: [data|process, parameters]
 Process Model: [process|parameters]
 Parameter Model: [parameters]

In general, Bayes' rule then gives the posterior distribution of the process and parameters given the data:

$$\begin{aligned} [\text{process, parameters}|\text{data}] &\propto [\text{data}|\text{process, parameters}] \\ &\times [\text{process}|\text{parameters}] \\ &\times [\text{parameters}]. \end{aligned}$$

In this general case, where the parameters are assigned prior distributions and hence are part of posterior inference, the model is referred to as a *Bayesian hierarchical model* (BHM).

In the case where the parameters are known or estimated from the data (traditionally known as *empirical Bayes*), the parameter model is dropped; we refer to this hierarchical model as an *empirical hierarchical model* (EHM). The general hierarchical model is discussed in Section 2.1; in this section, we are concerned with its use in Statistics for temporal data.

Data Models

A key idea behind the HM approach is that measurements of a process always occur with noise. Therefore, the parameter estimation presented in Sections 3.4.3–3.4.7 and 3.5.3–3.5.4 was under the ideal scenario of no measurement error. In general, it is not possible to observe the true state of a system. We might write this heuristically as follows: *Conditional* on the state process,

$$\text{observation process} = \text{state process} + \text{error process}, \quad (3.161)$$

where the state process describes the state of the system and is of primary interest. Because of the conditioning, the structure of the error process can be quite simple. Then we concentrate our (temporal) modeling effort on the state process at a lower level of the hierarchy. The advantages of this hierarchical modeling approach are discussed in Section 2.1.5.

Consider a data (or measurement) model often used in multivariate time series analysis:

$$\mathbf{Z}_t = \mathcal{H}(\mathbf{Y}_t) + \boldsymbol{\varepsilon}_t, \quad t = 0, 1, \dots, \quad (3.162)$$

where $\mathbf{Z}_t$ is an m_t -dimensional vector of observations at time t , $\mathbf{Y}_t$ is the corresponding n -dimensional vector representing the unobservable state process, $\mathcal{H}$ is a measurement function that relates the observation vector to the

state vector (often assumed to be known), and $\boldsymbol{\varepsilon}_t$ is the corresponding m_t -dimensional vector of observation errors, usually assumed to be independent in time. The classical data model assumes a *linear* measurement function $\mathcal{H}$ and independent Gaussian errors for $\{\boldsymbol{\varepsilon}_t\}$. That is,

$$\mathbf{Z}_t = \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t; \boldsymbol{\varepsilon}_t \sim \text{ind. Gau}(\mathbf{0}, \mathbf{R}_t),$$

where $\mathbf{H}_t$ is an $m_t \times n$ matrix (of parameters), and $\mathbf{R}_t$ is an $m_t \times m_t$ error covariance matrix. The matrix $\mathbf{R}_t$ is also a source of parameters in the measurement model. From Chapter 2, we call the collection of unknown parameters $\boldsymbol{\theta}_D$.

Process Models

Process models describe the underlying scientific process of interest, $\{Y_t\}$. They are also often factored further into a series of conditional models: Let $\mathbf{Y}_t = (Y_t^{(1)}, Y_t^{(2)})'$; $t = 0, 1, \dots$, denote a two-dimensional vector time series; then their joint probability can be written as

$$[Y_t^{(1)}, Y_t^{(2)} | \boldsymbol{\theta}_P] = [Y_t^{(1)} | Y_t^{(2)}, \boldsymbol{\theta}_P] [Y_t^{(2)} | \boldsymbol{\theta}_P].$$

For example, if $Y_t^{(1)}$ corresponds to a process Y at time t and $Y^{(2)}$ corresponds to the *same* process Y but at time $t - 1$, then this is just a model for Markovian evolution structure (e.g., AR(1)), as discussed in Section 3.4.7. In general, such factorizations are also important for simplifying multivariate processes. For example, Berliner, Milliff, and Wikle (2003) consider such a conditional framework for modeling the time-varying ocean conditioned on the time-varying atmosphere (see Section 7.4.3).

In so-called “state-space” modeling in time series analysis, a general evolution model for the process might be written as

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}) + \boldsymbol{\eta}_t, \quad (3.163)$$

where $\mathcal{M}$ is the transition function that describes the one-step evolution of the state process, and $\boldsymbol{\eta}_t$ is independent of $\mathbf{Y}_{t-1}$. A typical form of the model assumes this evolution function is *linear* (i.e., VAR(1)), such that $\mathbf{Y}_t = \mathbf{M}_t \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t$, where in general the $n \times n$ propagator (or transition) matrix $\mathbf{M}_t$ can be time-varying. Furthermore, the error process $\{\boldsymbol{\eta}_t\}$ is white noise in time and, typically, $\boldsymbol{\eta}_t \sim \text{Gau}(\mathbf{0}, \mathbf{Q}_t)$, where the covariance matrix $\mathbf{Q}_t$ can also be time-varying. We write this as $\boldsymbol{\eta}_t \sim \text{ind. Gau}(\mathbf{0}, \mathbf{Q}_t)$.

One can make various inferences on $\mathbf{Y}_t$, given the data. For observations $\mathbf{Z}_1, \dots, \mathbf{Z}_T$ and index $t \in \{1, \dots, T-1\}$, a *filtered* estimate of $\mathbf{Y}_t$ is given by $E(\mathbf{Y}_t | \mathbf{Z}_1, \dots, \mathbf{Z}_t)$; a *smoothed* estimate is given by $E(\mathbf{Y}_t | \mathbf{Z}_1, \dots, \mathbf{Z}_T)$; and a *forecast* estimate is given by $E(\mathbf{Y}_{T+\tau} | \mathbf{Z}_1, \dots, \mathbf{Z}_T)$, for $\tau > 0$. Section 8.2 can be consulted for further details on filtering, smoothing, and forecasting.

Parameter Models

In the case where the parameters in the data and process models are known we obviously do not need to specify parameter models. Similarly, in cases where these parameters are estimated from the data, the HM has no parameter model. This is an EHM and more akin to the traditional state-space estimation in time series. In contrast, under a BHM scenario, we specify models (distributions) for the various parameters appearing in the previous levels of hierarchy. Like before, parameter models from each stage can also be factored into subcomponents. For example, we often assume independence:

$$[\theta_D, \theta_P] = [\theta_D][\theta_P].$$

Furthermore, scientific insight and previous studies can facilitate the specification of $[\theta_D]$ and $[\theta_P]$. For example, measurement-error parameters in θ_D can often be obtained from previous studies that focus on such issues (this is typically the case for environmental variables and some ecological data). Certainly, process parameters in θ_P often carry scientific insight as well (e.g., Wikle et al., 2001; Wikle, 2003b).

In the specific data model (3.162) and process model (3.163) given above, if the parameters are known, then relatively computationally efficient estimates of the state process can be obtained if the functions $\mathcal{M}$ and $\mathcal{H}$ are linear and the error distributions are Gaussian. However, most commonly, the parameters are not known. Chapters 8 and 9 describe implementation and inference for EHMs and BHMs fitted to spatio-temporal data.

3.7 BIBLIOGRAPHIC NOTES

General Time Series

There are many excellent books on statistical time series analysis, including those for both the time domain and the frequency domain. These books provide a much more thorough background of the theory and application of time series than presented in this relatively brief overview chapter. A partial list of such books that we have found particularly helpful in writing this chapter are: Box and Jenkins (1970), Chatfield (1989), Brockwell and Davis (1991), Harvey (1993), Lütkepohl (1993, 2005), Fuller (1996), West and Harrison (1997), Bloomfield (2000), Shumway and Stoffer (2000, 2006), Durbin and Koopman (2001), Kitagawa (2010), and Prado and West (2011). Chandler and Scott (2010) present time series methods for trend detection and analysis, especially for environmental processes.

Dynamical Systems

A search of any academic library will provide many excellent references on dynamical systems, linear systems theory, and nonlinear dynamics. The discipline has progressed to the point where one can find such references at all

levels of technical sophistication. General texts that we have found particularly helpful are Sandefur (1990), Lorenz (1993), and Devaney (2003). For a statistical perspective, we highly recommend the paper by Bartlett (1990), the review paper by Berliner (1992), and the references therein, as well as the article by Wolff (1998) and the texts by Tong (1990) and Chan and Tong (2001). We have also found Caswell (2001) to be excellent in its discussion of multivariate linear systems. Some classical works in the area that we have found helpful include Lorenz (1963), Hénon (1976), Mandelbrot (1982), Takens (1981, 1985), and Gleick (1987). A classical text on random continuous-time processes is Srinivasan and Vasudevan (1971). The monographs by Majda, Abramov, and Grote (2005) and Majda and Wong (2006) provide a nice exposition of nonlinear dynamical systems for geophysical processes.

Nonlinear Time Series

Classical references for statistical nonlinear time series include Priestley (1988) and Tong (1990). These are invaluable introductions to a technically demanding subject that is growing in importance. A more-recent text with a nonparametric emphasis is Fan and Yao (2005). In addition, many general time series texts include at least a chapter on nonlinear aspects of time series; an excellent example is found in Chatfield (2004, Chapter 11). Furthermore, many of the general time series texts include discussion of GARCH and stochastic-volatility models. Books in financial econometrics also include a discussion of nonlinear time series (e.g., Tsay, 2005). Although not considered in this book, we note that there is also an extensive literature on neural network methods for analyzing nonlinear behavior in time series (e.g., Kay and Titterton, 2000).

Spectral Representations/Analysis

The basic background of spectral analysis and spectral representations can often be found in most of the texts on statistical time series analysis cited above. In addition, we have found the book by Walter (1994) to be an excellent source for general spectral decompositions, and the books by Marple (1987) and Percival and Walden (1993) are particularly helpful in applications of spectral analysis. In addition, we recommend the classic papers by Daniell (1946), Bartlett (1948), Cooley and Tukey (1965), Welch (1967), Thomson (1982), and Mallat (1989). We have also made reference to examples found in Wikle, Madden, and Chen (1997) and Wikle et al. (2001). Background material on equatorial waves and their observation can be found in Matsuno (1966) and Yanai and Maruyama (1966).

Hierarchical Statistical Methods

The notion of hierarchical statistical modeling has been around for quite some time (see the Bibliographic Note in Gelman et al., 2003, Section 5.7). A general discussion can also be found in Sections 2.1–2.3. Indeed, almost any modern text in Bayesian theory, methods, or applications will have a substantial

discussion of hierarchical statistical modeling. Gelman et al. (2003) is an excellent place to start. For an empirical hierarchical modeling (i.e., empirical Bayes) approach, see Carlin and Louis (2000). Banerjee, Carlin, and Gelfand (2004) discuss the BHM approach for spatial models.

The more modern texts in time series analysis contain material on hierarchical Bayesian analysis (e.g., West and Harrison, 1997; Shumway and Stoffer 2000, 2006). An important paper on Bayesian estimation of nonlinear time series is Carlin, Polson, and Stoffer (1992). Our perspective is shaped significantly by the paper of Berliner (1996). This approach is further outlined in the book by Royle and Dorazio (2008) as well as in papers by Wikle, Berliner and Cressie (1998), Berliner, Wikle, and Cressie (2000), Wikle et al. (2001) Berliner, Milliff, and Wikle (2003), and Wikle (2003a,b), among many others discussed here and in later chapters.

General References:

We have found excellent background material related to basis functions in Kaplan (1991). General background on statistical theory and matrix identities can be found in Rao (1965). Background on the theory of financial temporal processes can be found in Tsay (2005) and Wilmott, Howison, and Dewynne (1995).